Arithmetic structure of the standard genetic code

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Abstract

The standard genetic code is near-universal and stable over evolutionary time, motivating study of its arithmetic properties. We analyze proton and neutron counts of the encoded amino acids (free neutral forms) across 33 codon groups constructed from two natural grouping systems: the Rumer partition by family degeneracy and the three chemical axes of the third codon position (Keto/Amino, Strong/Weak, Purine/Pyrimidine). Excluding the three stop codons and the ATG initiator leaves 60 codons whose group sums determine an arithmetic structure modulo 37. Neutron counts reach the 37-lattice, whereas necessarily even proton counts, barred from odd multiples of 37, fall one below them: globally, and in the single tryptophan/isoleucine pair carrying the Keto/Amino asymmetry, with differences 37 and 36. The Keto/Amino balances and global conditions yield separate neutron and proton equations modulo 37, the former containing $111 = 3 \cdot 37$; since $\gcd(4, 37) = 1$, these equations fix a unique minimal nonnegative integer assignment for the four service codons, after which the regularities form a connected family of identities and divisibilities. Across 27 NCBI translation tables, this combination occurs only in the standard code and its exact codon-to-amino-acid equivalent; deterministic controls show divisibilities concentrated in third-position groupings, with 37 standing out among tested prime moduli. Under a random-code null, the observed values of several arithmetic features of this structure fall in the far tails of their Monte Carlo null distributions. Throughout, the regularities are properties of codon-group sums, not individual amino acids, and depend jointly on codon degeneracy and nucleon composition.

Contents

1 Introduction

The genetic code is one of the most ancient and conserved molecular features of cellular organisms. Its near-complete universality across the known domains of life, and the limited scope of variation in the known alternative codes, indicate that the basic structure of the standard code was already established at the early stages of biological evolution. Any organizational regularities present in its arithmetic structure therefore require precise analysis as one of the few potential sources of information about the early stages of life.

The arithmetic structure of the standard genetic code has been studied since shortly after its decipherment. Yu. B. Rumer [6, 7] introduced the partition of the 64 codons into two classes of 32 codons each, Octet I (the eight four-fold degenerate families) and Octet II (the remaining codons), related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$. Shcherbak and Makukov [8] observed that the number 37 plays a recurrent role in the arithmetic of the code, appearing as a divisor of amino acid nucleon sums across various groupings of codons. Panov and Filatov [5] revisited this analysis, introducing the per-codon counting rule, which forms the methodological basis of the present study.

The present analysis uses the proton and neutron counts of the twenty canonical amino acids in their free neutral forms under the most-abundant-isotope convention; once these conventions are fixed, the counts are uniquely determined. These quantities are considered together with two natural grouping systems defined independently of the nucleon arithmetic. The first is the Rumer partition. The second is given by the three chemical axes of the third codon position — Keto/Amino, Strong/Weak, and Purine/Pyrimidine — which exhaust the partitions of the four nucleotide bases into two pairs and correspond to the functional-group, hydrogen-bond, and ring-topology dichotomies of nucleobase chemistry.

The present work shows that within this framework the arithmetic regularities of the standard genetic code organize into a connected family of identities and divisibilities. The analysis traces this connected structure to a small number of empirical properties of the standard code. The same arithmetic criteria are then applied across all 27 NCBI translation tables. Among the tables analyzed, the tested combination of structural regularities occurs only in the standard table and in its exact codon-to-amino-acid equivalent. The specificity of the structure is examined further by deterministic controls, which test its dependence on the third codon position, on the modulus, and on the molecular representation of the amino acids, and by a Monte Carlo test against an explicit random-code null model.

The analysis is divided into three levels. The principal regularities are established at a parametrization-independent level, at which the three stop codons and ATG in its initiator role are excluded and only the remaining 60 codons are summed; these are the central results of the paper, and they involve no assigned values. To obtain sums for the complete table of 64 codons, however, the four service codons must be given some nucleon contributions. The second level is the naive reading (Key 0), in which the stop codons are assigned zero and ATG is counted as methionine. Assigning zero is also a choice of value, not the absence of one. The third level is the symmetrization (Key 1), in which the four service codons carry derived nucleon contributions. Once the choice of symmetry conditions is fixed, these contributions are determined as the unique minimal nonnegative integer solution of the corresponding system (Section 3), and the resulting regularities are consequences of that solution.

With regard to these assigned values, a methodological clarification is necessary, as it determines how the results should be interpreted. The assignment of values (P, N) to the service codons is a formal device intended to render the arithmetic of the complete table as a unified object, rather than a statement about any physical basis. Sums that include the service codons under such a parametrization are formal quantities introduced solely to reveal the underlying mathematical structure.

2 Methods

This section defines the nucleon parameters, the codon group systematics, and the parametrizations of service codons used throughout the analysis, together with the computed datasets supporting it.

2.1 Nucleon composition of amino acids

Throughout this work, each of the twenty canonical amino acids is represented as a free neutral amino acid molecule (not as a residue in a peptide chain), with its molecular formula retrieved from the PubChem compound database [3]. From this formula, we compute four integer quantities that characterize the nucleon content of the molecule.

The proton count P is defined from the elemental composition of the molecular formula as

$$P = \sum_E n_E Z_E,$$

where n_E is the number of atoms of element E in the formula and Z_E is its atomic number. The atomic numbers used are $Z(\mathrm{H}) = 1$, $Z(\mathrm{C}) = 6$, $Z(\mathrm{N}) = 7$, $Z(\mathrm{O}) = 8$ and $Z(\mathrm{S}) = 16$.

The neutron count N is computed by assigning each element its most abundant naturally occurring stable isotope, using the terrestrial isotopic compositions evaluated by IUPAC [1]. Accordingly, we use $^1\mathrm{H}$, $^{12}\mathrm{C}$, $^{14}\mathrm{N}$, $^{16}\mathrm{O}$ and $^{32}\mathrm{S}$. The total neutron count is

$$N = \sum_E n_E (A_E - Z_E),$$

where A_E is the mass number of the selected isotope of element E.

The total nucleon count T and the proton–neutron difference Δ are defined as

$$T = P + N, \qquad \Delta = P - N.$$

Both quantities are integers by construction. Under the isotope assumptions above, hydrogen is the only element contributing to Δ, since all other isotopes used have equal numbers of protons and neutrons. Consequently Δ is numerically equal to the number of hydrogen atoms in the molecule, and in particular $\Delta \geq 0$ for every amino acid.

To avoid notational collision, we use the uppercase symbol Δ exclusively for the proton–neutron imbalance of a single molecule or codon group, and reserve the lowercase symbol δ for pairwise differences between two such entities (for example, $\delta P = P_i - P_j$ between two amino acids i and j).

We emphasize that P, N, T and Δ are not measured molecular masses. They are exact integer quantities derived from elemental composition, and all arithmetic identities reported in this study are exact integer relations rather than artifacts of rounding approximate molecular masses. The sensitivity of the results to the choice of molecular representation — the free molecule or the amino acid residue in a peptide chain — is examined in Section 4.2.

The resulting nucleon values for all twenty canonical amino acids are provided in the dataset `amino_acids_nucleons.csv`, with one row per amino acid and columns for P, N, T and Δ. From this table we additionally compute three companion datasets of pairwise absolute differences — `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon_differences.csv` — which group all $\binom{20}{2} = 190$ unordered amino acid pairs by their values of $|\delta P|$, $|\delta N|$ and $|\delta T|$ respectively, where for any pair of amino acids (i, j) we write $\delta P = P_i - P_j$ and analogously for N and T. These companion datasets are used in later sections to identify amino acid pairs whose nucleon differences match specific target values, and in particular to locate pairs that are unique within the twenty amino acids with respect to a given difference.

All subsequent nucleon computations for codon groups reduce, by linearity, to sums of the per-amino-acid values tabulated in `amino_acids_nucleons.csv`, weighted by the multiplicities of the corresponding codons. Where service codons are included, their contributions are determined by the chosen parametrization (Section 2.4).

2.2 Genetic code structure

The standard genetic code defines a surjective mapping from the 64 codons of the four-letter nucleotide alphabet $\{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}$ to the twenty canonical amino acids and a translation-termination signal. Throughout this paper, we use DNA notation (with thymine T rather than uracil U) and adopt the codon assignments of The Standard Code (NCBI translation table 1) [4]. The three codons TAA, TAG and TGA act as stop signals, whereas ATG encodes methionine and serves as the canonical translation-initiation signal. For the purposes of the present arithmetic analysis, these four codons — the three stop codons and ATG in its initiator role — are referred to as *service codons.* Among possible initiation codons, ATG is predominant and is the only one whose RNA equivalent AUG is fully complementary to the CAU anticodon of the initiator methionine tRNA; the present analytical framework therefore uses ATG to represent the initiation function, which does not exclude initiation from other codons.

We define the *service-codon-excluded 60-codon pool* as

$$\mathcal{C}_{\mathrm{SE}} = \mathcal{C}_{\mathrm{standard}} \setminus \{\mathrm{ATG}, \mathrm{TAA}, \mathrm{TAG}, \mathrm{TGA}\},$$

where $\mathcal{C}_{\mathrm{standard}}$ denotes the set of all 64 codons of the standard code. Thus, $|\mathcal{C}_{\mathrm{SE}}| = 60$. Every codon belonging to $\mathcal{C}_{\mathrm{SE}}$ is counted according to its standard amino-acid assignment. The full codon-to-product mapping is provided in the dataset `genetic_code_codons.csv`.

By a codon family, we mean the set of four codons sharing the same first two bases and having an arbitrary third-position base, denoted by N. A structural property of the standard code, first identified by Rumer [6, 7], is the partition of the 64 codons into two halves of 32 codons each, distinguished by the degeneracy pattern of the third codon position. The first half, which we denote *Octet I*, consists of the eight codon families in which all four third-position nucleotides encode the same amino acid. These fully four-fold degenerate families are GCN (Ala), CGN (Arg), GGN (Gly), CCN (Pro), ACN (Thr), GTN (Val), CTN (Leu) and TCN (Ser). The second half, denoted *Octet II*, consists of the remaining eight codon families, in which the third-position nucleotide affects the encoded product. Every service codon of the standard code belongs to Octet II and carries a purine (A or G) at the third position.

The Rumer partition is not an analytical construct introduced in the present work but a structural feature of the code itself. It reflects both the qualitative distinction between fully and partially degenerate codon families and the quantitative fact that these two classes have equal cardinality: each contains eight families and thirty-two codons. Because Octet I contains no service codons, all nucleon quantities computed over Octet I are independent of any parametrization of start and stop signals and are determined solely by the nucleon content of the encoded amino acids. The same property holds, for a separate reason, for any codon group whose third-position nucleotide is restricted to pyrimidines (cytosine or thymine), since all four service codons of the standard code carry a purine at the third position. Full four-fold degeneracy and a purine-free third position are two distinct structural conditions, but both have the same consequence: nucleon quantities over the corresponding groups are fixed by their amino-acid composition and provide parametrization-independent reference values for the arithmetic identities of Section 3.

A further structural property of the Rumer partition, introduced by Rumer [7], is that the two octets are related by a simple permutation of the nucleotide alphabet. Under the transformation

$$R = (\mathrm{T} \leftrightarrow \mathrm{G},\ \mathrm{C} \leftrightarrow \mathrm{A}),$$

each codon family of Octet I is mapped to a codon family of Octet II, and vice versa. This transformation, known as the Rumer transformation, rearranges the partition but does not preserve the amino acids encoded by individual codons.

Taken together, the four-fold degeneracy pattern and the Rumer transformation identify the Octet I / Octet II partition as a natural structural division of the standard code, independent of the nucleon arithmetic. In the present work, this division is used as a reference partition over which nucleon quantities are computed and compared.

The codon table (Figure 1) is displayed with its axes in the base ordering C, G, T, A, which is also the default ordering of the accompanying visualizer. This ordering has two independent justifications. Chemically, it was introduced by Rumer [7] on the basis of base type and hydrogen-bond strength. The sequence C, G, T, A alternates pyrimidines and purines, while within each of these classes the member of the Strong pair $\{\mathrm{C}, \mathrm{G}\}$ precedes the member of the Weak pair $\{\mathrm{T}, \mathrm{A}\}$: C precedes T among the pyrimidines, and G precedes A among the purines. Geometrically, as shown by Panov and Filatov [5], it is one of exactly four orderings (out of 24 possible permutations of the four bases) in which Octet I and Octet II form connected regions on the codon matrix.

To assess the extent to which the arithmetic properties studied in this paper are specific to the standard genetic code, we additionally retrieved the full set of 27 translation tables maintained by NCBI [4]. The tables are numbered 1–33, with some numbers left unused, and cover nuclear, mitochondrial and plastid codes across a broad taxonomic range. The codon assignments of all these tables are collected in the dataset `ncbi_genetic_code_registry.csv` and are used in Section 4.1 to compare the standard code with the other NCBI translation tables.

2.3 Codon-group construction

In addition to the Rumer partition introduced in Section 2.2, we use a complementary overlapping grouping system based on the chemical identity of the nucleotide at the third codon position. Each of the four nucleotides belongs to one of two classes under each of three distinct binary chemical classifications:

- base type: *purine* $\{\mathrm{A}, \mathrm{G}\}$ (R) or *pyrimidine* $\{\mathrm{C}, \mathrm{T}\}$ (Y);

	C	G	T	A
C	CCC Pro 115 62 53	CGC Arg 174 94 80	CTC Leu 131 72 59	CAC His 155 82 73
	CCG Pro 115 62 53	CGG Arg 174 94 80	CTG Leu 131 72 59	CAG Gln 146 78 68
	CCT Pro 115 62 53	CGT Arg 174 94 80	CTT Leu 131 72 59	CAT His 155 82 73
	CCA Pro 115 62 53	CGA Arg 174 94 80	CTA Leu 131 72 59	CAA Gln 146 78 68
G	GCC Ala 89 48 41	GGC Gly 75 40 35	GTC Val 117 64 53	GAC Asp 133 70 63
	GCG Ala 89 48 41	GGG Gly 75 40 35	GTG Val 117 64 53	GAG Glu 147 78 69
	GCT Ala 89 48 41	GGT Gly 75 40 35	GTT Val 117 64 53	GAT Asp 133 70 63
	GCA Ala 89 48 41	GGA Gly 75 40 35	GTA Val 117 64 53	GAA Glu 147 78 69
T	TCC Ser 105 56 49	TGC Cys 121 64 57	TTC Phe 165 88 77	TAC Tyr 181 96 85
	TCG Ser 105 56 49	TGG Trp 204 108 96	TTG Leu 131 72 59	TAG STOP 0 0 0
	TCT Ser 105 56 49	TGT Cys 121 64 57	TTT Phe 165 88 77	TAT Tyr 181 96 85
	TCA Ser 105 56 49	TGA STOP 0 0 0	TTA Leu 131 72 59	TAA STOP 0 0 0
A	ACC Thr 119 64 55	AGC Ser 105 56 49	ATC Ile 131 72 59	AAC Asn 132 70 62
	ACG Thr 119 64 55	AGG Arg 174 94 80	ATG Met 149 80 69	AAG Lys 146 80 66
	ACT Thr 119 64 55	AGT Ser 105 56 49	ATT Ile 131 72 59	AAT Asn 132 70 62
	ACA Thr 119 64 55	AGA Arg 174 94 80	ATA Ile 131 72 59	AAA Lys 146 80 66

Figure 1: Standard genetic code with nucleon parameters of the encoded amino acids, displayed under the naive reading (Key 0), in which stop codons contribute zero and ATG is counted as methionine. In each cell, the codon appears on the left; the symbol of the encoded product (the three-letter amino acid symbol or STOP for the three stop codons) appears at the top center; the total nucleon count T appears below the product symbol; the proton count P appears at the top right; and the neutron count N appears at the bottom right. Codons of Octet I (the eight fully four-fold degenerate families) are shaded purple, and codons of Octet II are shaded yellow. Generated by the accompanying `visualization.html`.

- number of hydrogen bonds in the canonical base pair: *strong* $\{\mathrm{C},\mathrm{G}\}$ (S; three bonds) or *weak* $\{\mathrm{A},\mathrm{T}\}$ (W; two bonds);
- functional class: *amino* $\{\mathrm{A},\mathrm{C}\}$ (M) or *keto* $\{\mathrm{G},\mathrm{T}\}$ (K).

These axes were not selected in response to the subsequent nucleon calculations: they are three conventional pairwise dichotomies of the nucleotides and are represented explicitly in the IUPAC–IUB ambiguity nomenclature by the complementary symbol pairs R/Y, S/W, and M/K. Together they exhaust the three possible partitions of four nucleotides into two unordered pairs. For each resulting subset $X \subset \{\mathrm{A},\mathrm{T},\mathrm{G},\mathrm{C}\}$ with $|X| = 2$, we define the corresponding codon group as the set of all codons whose third-position nucleotide belongs to X.

For each of these six two-element subsets, and for each singleton $X \in \{\{\mathrm{A}\},\{\mathrm{C}\},\{\mathrm{G}\},\{\mathrm{T}\}\}$, we define the codon group

$$G_X = \{(c_1, c_2, c_3) \in \mathcal{C}_{\text{standard}} : c_3 \in X\}.$$

Thus, G_X consists of all codons whose third-position nucleotide belongs to X. The six two-nucleotide groups overlap: every codon belongs to exactly three of them, one for each chemical classification. At the same time, every codon belongs to exactly one of the four one-nucleotide groups.

This grouping system is applied at three levels: to all 64 codons, to the 32 codons of Octet I, and to the 32 codons of Octet II. If $|X| = 2$, then G_X contains 32 codons, while each of the intersections $G_X \cap$ Octet I and $G_X \cap$ Octet II contains 16 codons. If $|X| = 1$, the corresponding sizes are 16 and 8 codons. Group size alone does not identify a particular group; formulas therefore specify the set X and, where necessary, the octet explicitly.

The family of 33 codon groups used in the present work comprises the full code as one group of 64 codons; the two octets as groups of 32 codons each; ten groups G_X at the full-code level, corresponding to the six two-nucleotide and four one-nucleotide sets X; and their twenty intersections with Octet I and Octet II. Thus, the total number of analyzed groups is

$$1 + 2 + 10 + 20 = 33.$$

For example, $G_{\{\mathrm{G},\mathrm{T}\}} \cap$ Octet I is the 16-codon group Keto $\cap$ Octet I, whereas $G_{\{\mathrm{A}\}} \cap$ Octet II is the 8-codon group of Octet II with adenine at the third position. All 33 groups, together with their sizes, codon compositions, and Rumer-octet membership, are listed in the dataset `codon_groups.csv`.

A direct consequence of the structure of the standard code is the *twin-group property.* Within every codon family, the codons NNC and NNT with identical first two positions encode the same amino acid. Therefore, exchanging $\mathrm{C} \leftrightarrow \mathrm{T}$ at the third position does not change the multiset of encoded amino acids. Consequently, the Keto group $\{\mathrm{G},\mathrm{T}\}$ and the Strong group $\{\mathrm{C},\mathrm{G}\}$ have the same amino-acid composition, as do the Amino group $\{\mathrm{A},\mathrm{C}\}$ and the Weak group $\{\mathrm{A},\mathrm{T}\}$. The service codons also occur identically in the two groups of each pair. Hence, for any nucleon quantity $Q \in \{P, N, T, \Delta\}$,

$$Q(\text{Keto}) = Q(\text{Strong}), \qquad Q(\text{Amino}) = Q(\text{Weak})$$

under any common parametrization of the service codons. The same identities hold after intersecting each pair with Octet I or Octet II. We refer to these pairs of two-nucleotide groups as *twin groups*; in the dataset `codon_groups.csv`, the twin relationship is recorded in the column `Twin_Group_ID`.

A separate direct consequence of the same pyrimidine synonymy is the equality of the one-nucleotide groups $\{\mathrm{C}\}$ and $\{\mathrm{T}\}$ at the full-code level and after their intersection with either octet. At the full-code level, each contains 16 codons; within either octet, each contains 8 codons. Because they do not arise by comparing two branches of a chemical axis, the term "twin groups" and the column `Twin_Group_ID` are not used for them in the present work. Both groups and their intersections with the octets contain no service codons; hence, both their values and the equalities between them are independent of parametrization.

2.4 Parametrization of service codons

The nucleon quantities P, N, T and Δ of any codon group are obtained by summing the corresponding per-amino-acid values from Section 2.1 over all codons in the group. For the 60 sense codons (defined below), this summation is unambiguous: each such codon has a well-defined translation product with a well-defined molecular composition. The remaining four codons — the three stop signals TAA, TAG, TGA and the start signal ATG — require a separate convention, because stop codons do not correspond to any translation product and because the start codon plays a dual role as an initiation signal and as a methionine-encoding codon. We refer to these four codons collectively as *service codons*. The remaining 60 codons we call *sense codons*.[1] The start codon ATG encodes methionine and is a fully functional sense codon at any internal position; it is grouped with the service codons, and excluded from the sense pool, solely in its role as the universal initiation signal, not because it lacks a translation product. Only the ATG initiator is removed on these grounds; every other codon that encodes an amino acid is counted as a sense codon. The sense pool is thus defined as the 64 codons less the three stop codons and the ATG initiator, and its membership does not depend on the choice of parametric key.

To formalize this, we introduce the notion of a *parametric key*: a fixed assignment of nucleon values (P, N) to each of the four service codons. A parametric key specifies how service codons contribute to group sums; it affects only the four service codons, while the nucleon values of the 60 sense codons defined above are fixed by the amino acid composition of Section 2.1 and are unaffected by the choice of key. In the present study we employ two parametric keys, both of which are recorded in the dataset `key_parameters.csv` under the identifiers `Key 0` and `Key 1`.

The naive reading (Key 0). Under the zero parametrization, each of the three stop codons contributes $(P, N) = (0, 0)$, reflecting the fact that a stop codon is not translated into any amino acid and therefore has no molecular product to which nucleon values could be assigned. The start codon ATG is treated as an ordinary methionine codon, with the nucleon values of the free methionine molecule, $(P, N) = (80, 69)$. This convention isolates the arithmetic contribution of the amino-acid-coding portion of the code (with the start codon treated simply as methionine) and is the natural choice when one wishes to study the nucleon structure of the translation without introducing any ad hoc values for non-translated signals.

Symmetrization (Key 1). Under the symmetrization, each of the three stop codons is assigned $(P, N) = (37, 37)$ and the start codon ATG is assigned $(P, N) = (1, 0)$, treated as an initiation marker distinct from methionine. Unlike the zero parametrization, the symmetrization is not adopted by convention but derived analytically as the minimal nonnegative integer solution of a system of symmetry constraints imposed on the four nucleon quantities across the codon groups of Section 2.3. The full derivation, together with the system of constraints and its uniqueness properties, is presented in Section 3; in the present section the symmetrization is introduced purely as a named parametric key to be used in the datasets and the subsequent analysis.

A remark on terminology is in order. The methodological choice underlying the present work is to restrict attention to parameters whose definition is unambiguous and whose interpretation does not depend on any convention beyond the choice of service codon parametrization. The total nucleon count T, the proton count P, and the neutron count N of each amino acid correspond

[1] We note that the term *sense codon* is sometimes used in the literature for all 61 non-stop codons, including the ATG initiator. Throughout this paper it denotes the 60 codons remaining after the three stop codons and the ATG initiator are set aside; this usage is fixed in the accompanying datasets and visualizer.

to the actual nuclear composition of the free molecule, restricted to the most abundant stable isotopes (^{1}H, ^{12}C, ^{14}N, ^{16}O, ^{32}S). Once service codons are assigned non-zero contributions to P and N under a parametrization such as Key 1, sums formed over codon groups that include service codons no longer correspond to the nuclear composition of any physical ensemble of molecules. The values continue to be called proton, neutron, and total nucleon counts for consistency with the contributions of the canonical amino acids, but in such sums they are formal arithmetic quantities, and the question of their biochemical meaning does not arise.

The dataset `key_parameters.csv` additionally contains eight reference parametric keys (`Key 2`–`Key 9`), which correspond to various alternative conventions considered during the analysis. They are included for completeness and do not play the role of Key 1, which is derived rather than adopted; they are not used in any of the identities reported in this paper. In the main text we refer to the two principal keys by the descriptive names *zero parametrization* and *symmetrization* respectively, with the dataset identifiers given in parentheses where precision is required.

2.5 Analytical criteria and classification of equalities

The present study systematically examines the nucleon quantities P, N, T and Δ across all codon groups defined in Section 2.3, under each of the two parametric keys of Section 2.4, and records the set of arithmetic properties that these quantities exhibit. To keep the analysis transparent, we organize the search around a fixed hierarchy of criteria stated below.

Criterion 1: Divisibility by 37. For each nucleon quantity P, N, T, Δ of each codon group, we record whether the value is exactly divisible by 37. Divisibility by 37 is the primary criterion of the present work, continuing the line of investigation opened by Shcherbak and Makukov [8] and developed by Panov and Filatov [5]. Where divisibility holds, we additionally record whether it holds in a stricter form — that is, whether the quotient is itself divisible by further small factors. Divisibility by 37 corresponds to a residue of zero modulo 37; in some observations discussed in Section 3, small nonzero residues across structurally related groups are themselves structurally informative. Divisibility expressions for all four nucleon quantities of all codon groups are provided in the datasets `key0_divisibility-37.csv` and `key1_divisibility-37.csv`.

Criterion 2: Roundness in the decimal system. In parallel with Criterion 1, we record whether nucleon quantities are round numbers in the decimal system, that is, exact multiples of a power of ten. This criterion is motivated by the empirical observation that several nucleon quantities of the standard code are multiples of 100 and 1000 — in particular, the parametrization-independent totals over Octet I. Although roundness is stated in decimal notation, it reflects divisibility by the prime factors 2 and 5, which exist independently of the choice of notation; the systematic analysis of divisibility by these and the other primes, together with its relation to the modulus 37, is given in Section 4.2. Where roundness holds, we additionally record whether it holds in a stricter form, as a multiple of a higher power of ten.

Roundness in the decimal system and divisibility by 37 are arithmetically independent conditions, since $\gcd(10, 37) = 1$.

Criterion 3: Equalities between codon groups. For each pair of nucleon quantities taken over two codon groups G_1 and G_2, we record whether the two quantities coincide numerically. Every such equality is classified according to the following three-tier scheme, which depends on the structural type of the equality:

- *Different-size equalities* — the equality relates nucleon quantities of two groups of *different* sizes (for example, a size-64 group and a size-32 group), regardless of whether the two quantities involved are of the same type or of different types. This is the strongest type of equality, because it reflects a coincidence between structurally independent sums computed over codon sets of different scale.

- *Cross-parameter equalities* — the equality relates two *different* nucleon quantities (for example, P of one group and N of another) belonging to groups of the *same* size. This is the intermediate type: the coinciding sums range over codon sets of equal cardinality, but the quantities being compared are structurally distinct.

- *Same-parameter equalities* — the equality relates the *same* nucleon quantity of two distinct groups of the same size (for example, $P(G_1) = P(G_2)$, where G_1 and G_2 have equal numbers of codons). This is the weakest type, since the two sums are of the same structural nature — the same quantity over codon sets of the same cardinality — and coincidence between them is the least informative form of equality. The twin-group identities of Section 2.3 are a special case of this type; they are recorded but treated as structural consequences of Section 2.3 rather than as emergent coincidences.

The classification depends only on the sizes of the compared groups and on the identity of the compared quantities, and not on the specific groups involved. All observed equalities for both parametric keys are tabulated in the datasets `key0_equalities.csv` and `key1_equalities.csv`, in which the three classes are encoded in the `Match_Type` column with values `DIFF_PARAM`, `CROSS_PARAM`, and `SAME_PARAM` respectively. Under Key 0 only same-parameter equalities occur; the stronger different-size and cross-parameter equalities arise only under Key 1.

Criterion 4: Additional structural observations. Beyond divisibility and equalities, we record several further types of structural regularity that do not fit into Criteria 1–3 but nevertheless carry information about the arithmetic organization of the code. These include:

- exact powers — in particular, powers of two or other small integers — realized by nucleon quantities of codon groups;

- anomalously simple reduced fractions arising as ratios P/T, N/T or P/N of nucleon quantities within a single codon group, tabulated in `key0_ratios.csv` and `key1_ratios.csv`;

- coincidences at the level of individual amino acid pairs, in which a nucleon difference $|\delta P|$, $|\delta N|$ or $|\delta T|$ between two amino acids matches a value that appears separately in the group-level analysis (for example, the Trp–Ile pair, for which $|\delta N| = 37$ coincides exactly with the divisor of Criterion 1).

The observations accumulated under Criteria 1–4 are presented in Section 3. They are grouped into parametrization-independent results, which hold for every parametric key, and parametrization-specific results, which depend on the choice of key.

2.6 Computed datasets and reproducibility

The core results reported in the present study are produced by a single reproduction script, `reproduce.py`, that consumes the five input datasets introduced in the previous sections and writes a fixed set of computed output datasets. Two further scripts, the control script `controls.py` and the Monte Carlo script `mc_significance.py` (with its independent re-check `verify_mc.py`), are described separately below and produce the supplementary datasets. This section summarizes the organization of the computed datasets and the reproducibility of the pipeline.

Input datasets. The five input datasets, all of which have been introduced in their respective sections, are: `amino_acids_nucleons.csv` (Section 2.1), `genetic_code_codons.csv` (Section 2.2), `ncbi_genetic_code_registry.csv` (Section 2.2), `codon_groups.csv` (Section 2.3), and `key_parameters.csv` (Section 2.4). These five files fully specify the nucleon composition of the amino acids, the codon assignments of the standard and alternative genetic codes, the partitioning of the 64 codons into groups, and the parametric keys used to handle service codons.

Per-key computed datasets. For each of the two principal parametric keys (Key 0 and Key 1), the pipeline produces four computed datasets with a uniform naming convention, where the prefix `key0_` or `key1_` indicates the active parametric key:

- `*_nucleon-data.csv` — the four fundamental nucleon quantities P, N, T, Δ computed for each of the 33 codon groups defined in `codon_groups.csv` under the active key;
- `*_divisibility-37.csv` — a compact expression of each of the above values as $q \cdot 37 + r$, making divisibility by 37 and residue modulo 37 immediately visible for every group;
- `*_equalities.csv` — the complete enumeration of all numerical equalities between nucleon quantities across codon groups, classified by the `Match_Type` scheme of Section 2.5 into `DIFF_PARAM`, `CROSS_PARAM` and `SAME_PARAM`;
- `*_ratios.csv` — pairwise ratios of the four nucleon quantities for each codon group, provided in both exact fractional form (reduced to lowest terms) and decimal approximation, used for locating anomalously simple fractions as described in Section 2.5.

Together, these eight per-key files contain every numerical value on which the results of Section 3 are based.

Key-independent computed datasets. Three further computed datasets are produced once and do not depend on the choice of parametric key: `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon_differences.csv`. These enumerate, for all 190 pairs of the twenty canonical amino acids, the absolute differences $|\delta P|$, $|\delta N|$ and $|\delta T|$ respectively, and are used to locate coincidences at the level of individual amino acid pairs as described in Section 2.5.

Sense-pool arithmetic across NCBI codes. The computed dataset `deficit_models_analysis.csv` extends the analysis beyond the standard code by applying the same nucleon bookkeeping to every translation table listed in `ncbi_genetic_code_registry.csv`. The standard code itself appears as one row of the dataset, serving as the reference point against which the remaining 26 tables are compared.

For each translation table the dataset records the global nucleon pools $N(\text{All})$ and $P(\text{All})$ and the sense pools $N(\text{All sense})$ and $P(\text{All sense})$ (total counts over the 60 sense codons of Section 2.4, that is, excluding the three stop codons and the ATG initiator). Two arithmetic tests are then tabulated, each with its own reference structure.

The first test records the deficit of the sense pool neutron pool relative to three reference anchors derived from Octet I. The total nucleon count $T(\text{Octet I})$ serving as each anchor is evaluated under two definitions: a *structural* definition (the eight codon families occupying the Octet I positions in the standard code, regardless of their amino acid assignments in the alternative code) and a *functional* definition (the codon families that are fully four-fold degenerate in the alternative code itself). The three anchors are the constant 3700, the structural $T(\text{Octet I})$ and the functional $T(\text{Octet I})$. For the standard code all three definitions coincide and the deficit equals $111 = 3 \cdot 37$ in all three columns (Section 3.3.1).

The second test records the residue of the sense pool proton pool $P(\text{All sense})$ modulo 37, without reference to any Octet I-based anchor. For the standard code this residue is $-1 \equiv 36 \pmod{37}$, the parity-induced unit shift discussed in Sections 3.1.1 and 3.1.5.

The alternative codes are evaluated against the same two criteria, and the results are used in Section 4.1 to assess the extent to which the arithmetic structure of the standard code is shared with its natural variants.

A secondary model, in which the context-dependent stop codons of the three tables 27, 28, and 31 (Karyorelict, Condylostoma, and Blastocrithidia Nuclear Codes) are treated as sense codons rather than as stops, is tabulated in parallel in the `_Alt` columns of the same dataset.

Keto/Amino balance across NCBI codes. The computed dataset `keto_amino_balance_models.csv` extends the Keto/Amino balance analysis of Section 3.1.5 beyond the standard code by applying the same nucleon bookkeeping to every translation table listed in `ncbi_genetic_code_registry.csv`. The standard code itself appears as one row of the dataset, serving as the reference point against which the remaining 26 tables are compared.

For each translation table, the dataset records the sense pool neutron and proton counts of the Keto and Amino half-codes, $N(\text{Keto sense})$, $N(\text{Amino sense})$, $P(\text{Keto sense})$, $P(\text{Amino sense})$, together with their pairwise differences $N(\text{Keto sense}) - N(\text{Amino sense})$ and $P(\text{Keto sense}) - P(\text{Amino sense})$. For the standard code these differences are 37 and 36 respectively (Section 3.1.5); the alternative codes are evaluated against the same pair of reference values, and the results are used in Section 4.1 to identify codes that reproduce the Keto/Amino signature of the standard code.

The same secondary sense-pool model as in the preceding paragraph — in which the context-dependent stop codons of Tables 27, 28, and 31 are returned to the sense pool — is tabulated in parallel in the `_Alt` columns. For the standard code, which contains no context-dependent stop codons, the primary and alternative columns coincide.

Reproducibility. The complete pipeline — input datasets, Python scripts, and all computed output datasets — is provided as a single repository accompanying this paper. Running the script (standard-library Python only, no external dependencies) on the five input datasets regenerates every computed dataset bit-for-bit, and every numerical value appearing in Section 3 can be traced directly to a specific cell of one of these datasets. Within the core pipeline, no randomization is used at any stage of the computation, and all arithmetic underlying the reported regularities is exact, carried out on integer-valued tables and, for ratios, on exact reduced fractions. Statistical significance is assessed separately by a fixed-seed Monte Carlo script; its tail probability estimates and uncertainty intervals are necessarily floating-point quantities. Within the deterministic core pipeline, the only floating-point values are the rounded decimal approximations printed alongside the exact fractions in the `*_ratios.csv` files, for readability; no result depends on them.

Supplementary visualization tool. An interactive HTML visualization of the genetic code table and its codon-group structure, used during the exploratory phase of this study, is included in the accompanying repository as a supplementary tool. The visualization is not part of the reproducible pipeline described above; any numerical value displayed in its interface under Key 0 or Key 1 that appears in the main text can be verified against the corresponding cell of the computed datasets. An interactive version of the visualizer is available online at `https://ruslankhafizov.github.io/genetic-code-arithmetic/`.

Supplementary control datasets. Three further datasets probe the specificity of the arithmetic structure and are produced by a separate script `controls.py`, which consumes the same input datasets but is not part of the core pipeline described above. Like that pipeline it uses standard-library Python only, exact integer arithmetic, and no randomization. The dataset `position_axis_analysis.csv` records the four nucleon quantities of six groups — the two sides of each of the three chemical axes — applied independently to the first, second, and third codon position, under the sense pool, Key 0, and Key 1; it is used in Section 4.2 to show that divisibility modulo 37 is concentrated at the third position. The dataset `prime_divisibility_scan.csv` records, for every prime up to 97 and for three value sets — the parametrization-independent groups, Key 0, and Key 1 — how many of the group quantities it divides, over all instances, over distinct values, and over distinct odd cores (the odd part of a number after removing factors of 2). It is used in the same section to show that 37 carries the largest excess apart from 2 and 5. The excess at 2 reflects proton parity, and that at 5 is a trace of the round-number regularity of Criterion 2 (divisibility by 10, 100, and 1000). The multiple distinct odd cores occur among the multiples of 37, whereas the smaller excesses at $73 = 2 \cdot 37 - 1$ and at 61 each reduce to a single odd core and are satellites of an individual quantity. This separation persists even in the value set in which divisibility by 37 is not imposed. The dataset `position_asymmetry_ncbi.csv` records, for each of the 27 NCBI translation tables, the Keto/Amino sense-pool asymmetry for the four nucleon quantities at each codon position, with residues modulo 37; it is used in Section 4.2 to show that this asymmetry is divisible by 37 at all three positions only for the standard code and its exact codon-to-amino-acid equivalent.

Data and code availability. The complete package supporting this work — the LaTeX source of the present preprint, the five input datasets, the thirteen computed datasets, the reproduction script `reproduce.py`, the supplementary control script `controls.py` with its three control datasets, the Monte Carlo script `mc_significance.py` with its summary table and independent re-check `verify_mc.py`, and the interactive visualizer — is available in the accompanying repository at `https://github.com/RuslanKhafizov/genetic-code-arithmetic` and is archived at Zenodo: DOI: 10.5281/zenodo.20360037.

3 Results

The arithmetic properties reported here separate into three levels. The first (Section 3.1) consists of facts that hold independently of any assignment to the service codons: sums over the sixty sense codons and over the codon groups that contain no service codon. These parametrization-independent facts are the central result of this work; their atypicality under an explicit random-code null is established in Section 4.3. The second level (Section 3.2) is the naive reading of the code, in which the three stop codons carry no nucleons and the initiator is counted as methionine; much of the same structure is already visible here, before any derived parametrization. The third level (Sections 3.3–3.5) introduces the unique minimal integer assignment to the service codons, Key 1, that extends the structure to the full 64-codon table. Key 1 is a device for exhibiting the arithmetic of the complete table, not a foundation on which the parametrization-independent findings rest.

The standard code is the principal object of analysis throughout; alternative NCBI translation tables are introduced later, for comparison and to assess the specificity of the results.

3.1 Parametrization-independent results

The facts collected in this subsection require no assignment to the service codons and are, in that sense, the most secure results of the paper. The most consequential of them is a property of the sense pool as a whole (Section 3.1.1). Beyond it, two categories of codon group contain no service codons at all—Octet I with all its sub-groups, and groups whose third-position nucleotide is restricted to pyrimidines—and are therefore parametrization-independent directly, while a further family of equalities between twin pairs of groups holds under any key even when the paired groups themselves contain service codons. More broadly, once the four service codons are excluded every group reduces to a sum over its remaining sense codons; Figure 2 shows these sense-pool sums across all three levels of partitioning (the grid layout is shared with Figures 3 and 4, and its conventions are given in the caption of Figure 3).

3.1.1 The sense pool: neutron divisibility and the proton residue

Summed over all sixty sense codons—the full code excluding the three stop codons and the initiator ATG—the neutron and proton totals are

$$N(\text{All sense}) = 3589 = 97 \cdot 37, \qquad P(\text{All sense}) = 4180 \equiv -1 \pmod{37}.$$

The neutron sum is an exact multiple of the modulus 37; the proton sum falls one unit short of one. This is the single most consequential empirical fact established here, and it is new: prior analyses of the code's nucleon arithmetic worked with the total count $T = P + N$ alone, whereas the divisibility of the neutron pool and the one-unit deficit of the proton pool are properties of the proton/neutron split, invisible at the level of T.

The direction of the proton deficit is fixed by a parity constraint. Every canonical amino acid is a neutral, closed-shell molecule and therefore contains an even number of protons (Section 3.1.5); a sum of even proton counts is itself even, and an even number cannot equal an odd multiple of 37 such as $113 \cdot 37 = 4181$. The proton pool cannot occupy that lattice point; its value, 4180, lies one unit below it. The neutron pool is subject to no such constraint and meets the lattice exactly. This asymmetry between the two registers—the neutron sum on the lattice, the proton sum one unit below—recurs throughout the results and is taken up in Section 3.5.

Whether a concentration of this kind is atypical is addressed in Section 4.3, where the standard code is compared against an explicit random-code null; the neutron divisibility reported here is one of the two anchors of that test.

3.1.2 Arithmetic of Octet I

The four nucleon quantities of Octet I (id 12) are

$$T(\text{Octet I}) = 3700, \qquad P(\text{Octet I}) = 2000,$$

$$N(\text{Octet I}) = 1700, \qquad \Delta(\text{Octet I}) = 300.$$

Notably, all four values are exact multiples of 100, with $P(\text{Octet I})$ additionally being an exact multiple of 1000. Beyond roundness, the total nucleon count $T(\text{Octet I}) = 3700 = 100 \cdot 37$ is also a multiple of 37. Its neutron content combines with the Octet II sense-pool neutron content to give the sense-pool neutron sum of Section 3.1.1,

$$N(\text{All sense}) = N(\text{Octet I}) + N(\text{Octet II sense}) = 1700 + 1889 = 3589 = 97 \cdot 37,$$

in which neither octet contribution is individually a multiple of 37. The Octet I total $T(\text{Octet I}) = 3700$ exceeds this sense-pool neutron sum by exactly $3 \cdot 37$; this identity underlies the neutron divisibility that fixes Key 1 (Section 3.3).

Because every codon family in Octet I is fully four-fold degenerate, the four values above propagate identically to all sub-groups of Octet I: each of the six groups of 16 codons within Octet I (ids 13–18) carries

$$T = 1850, \qquad P = 1000, \qquad N = 850, \qquad \Delta = 150,$$

and each of the four groups of 8 codons within Octet I (ids 19–22) carries

$$T = 925, \qquad P = 500, \qquad N = 425, \qquad \Delta = 75.$$

Divisibility by 37 of T propagates to these levels as well: $1850 = 50 \cdot 37$ and $925 = 25 \cdot 37$, so $T(\text{Octet I})$ is a multiple of 37 at every level of the Octet I hierarchy.

Beyond divisibility of T, the proton and neutron quantities at the 16-codon level of Octet I satisfy $P \equiv +1$ and $N \equiv -1 \pmod{37}$, a systematic ± 1 deviation compensating to zero in

Figure 2: Structure of the 60-codon service-codon-excluded pool (POOL60). Each group is summed over its non-service codons only; the three stop codons and the initiator ATG are excluded, so every quantity shown is parametrization-independent. (ATG is set aside only in its role as the initiation signal, not because it lacks a translation product.) For each group the four quantities are the total nucleon count T, the proton sum P, the neutron sum N, and the hydrogen count $\Delta = P - N$; green highlights mark divisibility by 37. Two distinct one-unit effects appear in the hydrogen count Δ. The first is the pyrimidine cascade of Section 3.1.4: the eight-codon Octet I boxes lie at $\Delta = 75 = 2 \cdot 37 + 1$ and the Octet II boxes at $\Delta = 73 = 2 \cdot 37 - 1$. Superimposed on it is a one-unit deficit confined to the groups with third-position G: the box $\{G\} \cap$ Octet II lies one below its Octet II siblings at $\Delta = 72$, which carries up to $\Delta(\{G\}) = 147 = 4 \cdot 37 - 1$ (against $148 = 4 \cdot 37$ for $\{A\}$, $\{C\}$, $\{T\}$) and to $\Delta = 295$ on the Keto, Strong and Purine axes (against $296 = 8 \cdot 37$ on Amino, Weak, Pyrimidine). The two effects are of the same magnitude but distinct in origin. The neutron side is aligned with the lattice: N is a multiple of 37 on Keto, Amino, Strong and Weak ($N = 1813 = 49 \cdot 37$ and $N = 1776 = 48 \cdot 37$), and consequently over the whole sense pool $N = 3589 = 97 \cdot 37$. Since N is aligned on all four of these axes, the proton sum $P = N + \Delta$ follows Δ: on Amino and Weak, where Δ lies on the lattice, P does too, whereas on Keto and Strong, where Δ is one unit short, P inherits the same shift, $P \equiv -1 \pmod{37}$.

$T = P + N$. At the 8-codon level, $\Delta = 75 = 2 \cdot 37 + 1$ shows the same +1 deviation directly. This pattern recurs in the pyrimidine-restricted groups of Section 3.1.4 and in parametrization-specific observations of Section 3.2.

The ratios P/T, N/T and Δ/T of Octet I reduce to anomalously simple fractions sharing a common denominator of 37:

$$\frac{P}{T} = \frac{20}{37}, \qquad \frac{N}{T} = \frac{17}{37}, \qquad \frac{\Delta}{T} = \frac{3}{37},$$

and, since every sub-group of Octet I consists of the same four-fold degenerate families in equal proportion, these ratios hold identically at every level of the hierarchy.

3.1.3 Pyrimidine synonymy and twin-group identities

In the standard genetic code, the codons NNC and NNT (where C and T are pyrimidine bases) encode the same amino acid.

A direct algebraic consequence of pyrimidine synonymy is the equality of nucleon quantities between the groups **Keto** $\{\mathrm{G}, \mathrm{T}\}$ (id 2) and **Strong** $\{\mathrm{C}, \mathrm{G}\}$ (id 3), and between **Amino** $\{\mathrm{A}, \mathrm{C}\}$ (id 4) and **Weak** $\{\mathrm{A}, \mathrm{T}\}$ (id 5). Each of these four groups contains 32 codons. The distribution of service codons across them is uneven: Keto and Strong contain the stop codon TAG and the start codon ATG, while Amino and Weak contain two stop codons, TAA and TGA, and no start codon. Each of the four groups thus contains 30 sense codons. Although these groups themselves contain service codons — so their individual nucleon values depend on the parametric key — the service codons enter both groups of each twin pair symmetrically, and the equality of nucleon quantities holds under any parametric key.

Analogous equalities hold between Keto $\cap$ Octet II and Strong $\cap$ Octet II (id 24 and id 25), between Amino $\cap$ Octet II and Weak $\cap$ Octet II (id 26 and id 27), and between the corresponding pairs within Octet I (id 13 and id 14, id 15 and id 16). All these twin pairs are recorded in the `Twin_Group_ID` column of `codon_groups.csv`. Within Octet I, full four-fold degeneracy forces all six 16-codon sub-groups to be mathematically identical, but only the Keto/Strong and Amino/Weak pairs are recorded in `Twin_Group_ID`. The Purine/Pyrimidine pair is excluded because outside Octet I the two groups have substantially different amino acid composition: Purine contains all four service codons, while Pyrimidine contains none. Their equality within Octet I is a degeneracy-specific artifact, not a manifestation of pyrimidine synonymy.

A separate phenomenon of the same origin is the equality of the single-nucleotide groups $\{\mathrm{C}\}$ (id 8) and $\{\mathrm{T}\}$ (id 10), together with their intersections with each octet. This equality is qualitatively different: it does not relate groups across a chemical axis of partitioning, but follows directly from pyrimidine synonymy itself — a codon with C at the third position and the same codon with T encode the same amino acid by definition. For this reason the single-nucleotide pairs $\{\mathrm{C}\}/\{\mathrm{T}\}$ are not recorded in `Twin_Group_ID`, which is reserved for pairs whose identical amino acid composition arises from pyrimidine synonymy across a two-nucleotide chemical axis (Keto/Strong or Amino/Weak). $\{\mathrm{C}\}$ and $\{\mathrm{T}\}$ form the only such pair in the standard code that is entirely free of service codons, and is therefore parametrization-independent both in its values and in their equality.

3.1.4 Arithmetic of pyrimidine-restricted groups

The Pyrimidine group (id 7) and its sub-groups form the second class of parametrization-independent groups of the standard code. Unlike Octet I, these groups span both octets and contain partially degenerate codon families; their arithmetic is therefore not inherited from a single set of values and must be examined separately at each level.

The full pyrimidine group (id 7). The nucleon quantities are

$$T = 4096, \qquad P = 2196, \qquad N = 1900, \qquad \Delta = 296.$$

Three of these four values simultaneously satisfy distinct criteria: $T = 4096 = 2^{12}$ is an exact power of two; $N = 1900$ is an exact multiple of 100; and $\Delta = 296 = 8 \cdot 37$ is divisible by 37.

At the 16-codon level, the groups {C} (id 8) and {T} (id 10) partition Pyrimidine into two identical halves with values $T = 2048 = 2^{11}$, $P = 1098$, $N = 950$, and $\Delta = 148 = 4 \cdot 37$.

Pyrimidine-restricted groups at the 8-codon level. Intersecting {C} and {T} with each octet yields four groups of 8 codons whose Δ values, under any parametric key, are

$$\Delta = 75 = 2 \cdot 37 + 1 \quad \text{for } \{\mathrm{C}\} \cap \text{Octet I (id 19) and } \{\mathrm{T}\} \cap \text{Octet I (id 21)},$$

$$\Delta = 73 = 2 \cdot 37 - 1 \quad \text{for } \{\mathrm{C}\} \cap \text{Octet II (id 30) and } \{\mathrm{T}\} \cap \text{Octet II (id 32)}.$$

These four values exhibit a systematic ± 1 deviation from a multiple of 37: the Octet I values lie one unit above $74 = 2 \cdot 37$, while the Octet II values lie one unit below. Adding the two 8-codon halves {C} and {T} within each octet recovers the intersections of the full pyrimidine group with the corresponding octet:

$$\Delta(\text{Pyrimidine} \cap \text{Octet I}) = 150 = 4 \cdot 37 + 2 \quad \text{(id 18)},$$

$$\Delta(\text{Pyrimidine} \cap \text{Octet II}) = 146 = 4 \cdot 37 - 2 \quad \text{(id 29)}.$$

The ± 1 deviations at the 8-codon level double to ± 2 at the 16-codon level within each octet. Adding across both octets recovers $\Delta(\text{Pyrimidine}) = 296 = 8 \cdot 37$, with the deviations of opposite sign cancelling exactly. The resulting cascade $\pm 1 \to \pm 2 \to 0$ — a systematic small deviation at the 8-codon level that doubles upon intra-octet combination and cancels upon inter-octet combination — extends to further groups under Key 0 (Section 3.2), where analogous one-unit deviations also appear in the proton and neutron residues.

3.1.5 Localization of the Keto/Amino asymmetry, and proton parity

Over the sense pool—parametrization-independent by construction—the differences between the Keto and Amino nucleon sums are

$$\begin{aligned}
T(\text{Keto sense}) - T(\text{Amino sense}) &= 3921 - 3848 = 73,\\
P(\text{Keto sense}) - P(\text{Amino sense}) &= 2108 - 2072 = 36,\\
N(\text{Keto sense}) - N(\text{Amino sense}) &= 1813 - 1776 = 37,\\
\Delta(\text{Keto sense}) - \Delta(\text{Amino sense}) &= 295 - 296 = -1,
\end{aligned}$$

where $X(G \text{ sense})$ denotes the sum of $X \in \{T, P, N, \Delta\}$ over the sense codons of the group G. These differences reduce to the nucleon counts of a single pair of molecules, and the reduction is structural.

Reduction to a single pair. Of the 16 codon families, 13 contain no service codon, and in each the codons NNG (Keto branch) and NNA (Amino branch) encode the same amino acid, contributing equally to both sums. The family TA also contributes equally: its pyrimidine pair TAC/TAT encodes Tyr in both branches, and its purine pair TAG/TAA consists of two stop codons, one in each branch, excluded from the sense pool. Asymmetry is realized only in the families AT (with ATA = Ile in the Amino branch; the Keto codon ATG is the excluded initiator) and TG (with TGG = Trp in the Keto branch; the Amino codon TGA is an excluded stop). In

TG, Cys is present symmetrically and does not contribute; in AT, the remaining Ile codons ATT and ATC are likewise symmetric. The sense-pool asymmetry therefore reduces to the differences of the pair Trp–Ile, that is $(73, 36, 37, -1)$. This localization holds for any assignment giving equal values to the three stop codons and depends only on the code's synonymy structure. (Under the naive reading, Key 0, the same asymmetry appears over the full code with the initiator's product included, $T(\text{Keto}) - T(\text{Amino}) = 222 = T(\text{Trp}) + T(\text{Met}) - T(\text{Ile})$.)

The pair and the modulus. The two nucleon differences that matter for what follows are $|\delta N(\text{Trp}, \text{Ile})| = 37$, equal to the modulus, and $|\delta P(\text{Trp}, \text{Ile})| = 36$, one unit below it. Among the 190 pairs of canonical amino acids, $|\delta N| = 37$ is realized only by Trp–Ile and Trp–Leu, and $|\delta P| = 36$ only by the same two; Leu gives the same values as Ile because the two are isomers with identical counts $P = 72$, $N = 59$, so a single nucleon profile is involved. The proton value follows from the neutron value and the hydrogen difference, $\delta P = \delta N - (\Delta_{\text{Ile}} - \Delta_{\text{Trp}}) = 37 - 1 = 36$, one below because isoleucine carries one more hydrogen than tryptophan.

These differences are properties of the two molecules alone and carry no independent statistical weight: the neutron difference coincides with the modulus at roughly the rate expected by chance among amino-acid pairs. They enter the derivation of Section 3.3 as the offsets that the service-codon balance must reproduce, not as evidence in themselves; the atypicality of the arithmetic resides in the aggregate (Section 4.3), not in this pair. The induced total difference $|\delta T| = |\delta P| + |\delta N| = 73 = 2 \cdot 37 - 1$ is not independent. Full distributions of $|\delta N|$, $|\delta P|$, and $|\delta T|$ over the 190 pairs are given in `amino_acids_neutron_differences.csv`, `amino_acids_proton_differences.csv`, and `amino_acids_nucleon_differences.csv`.

Proton parity. That $|\delta P|$ is even—so that it cannot equal the odd value 37 and sits at 36 rather than on the modulus—reflects a general parity constraint on the proton counts. Every one of the 20 canonical amino acids has an even proton count: each value of P in `amino_acids_nucleons.csv` is even. The reason is electronic. In their neutral ground states the canonical amino acids are closed-shell molecules and therefore contain an even number of electrons; since a neutral molecule has equal numbers of electrons and protons, its proton count is even. Consequently every proton sum or pairwise difference over the amino acids is even, and no such quantity can equal an odd multiple of 37. This is the constraint behind the one-unit proton deficit of the sense pool (Section 3.1.1) and behind $|\delta P| = 36$ here; parity fixes that these quantities miss the odd multiples of 37, while the size of the miss—one unit in both cases—is a property of the specific nucleon counts.

3.2 Results under the zero parametrization

Key 0 is the naive reading of the code: the three stop codons, which have no translation product, are assigned no nucleons, and the initiator ATG is counted as its product methionine. It introduces no derived parameters. We call it naive rather than physical because assigning zero to the stop codons is itself an assignment of values, not the absence of one; the genuinely assignment-free level is the sense pool of Section 3.1, over which the service codons are not summed at all. The properties below show that much of the arithmetic structure is already present at this reading, before the minimal parametrization of Section 3.3 is introduced—indeed the concentration on the lattice of step 37 is already atypical here (statistic S4a of Section 4.3). This bears on how the derived parametrization should be read: the structure is not created by assigning values to the service codons, only sharpened by it. Figure 3 gives a graphical overview of the quantities computed under Key 0 that are discussed below.

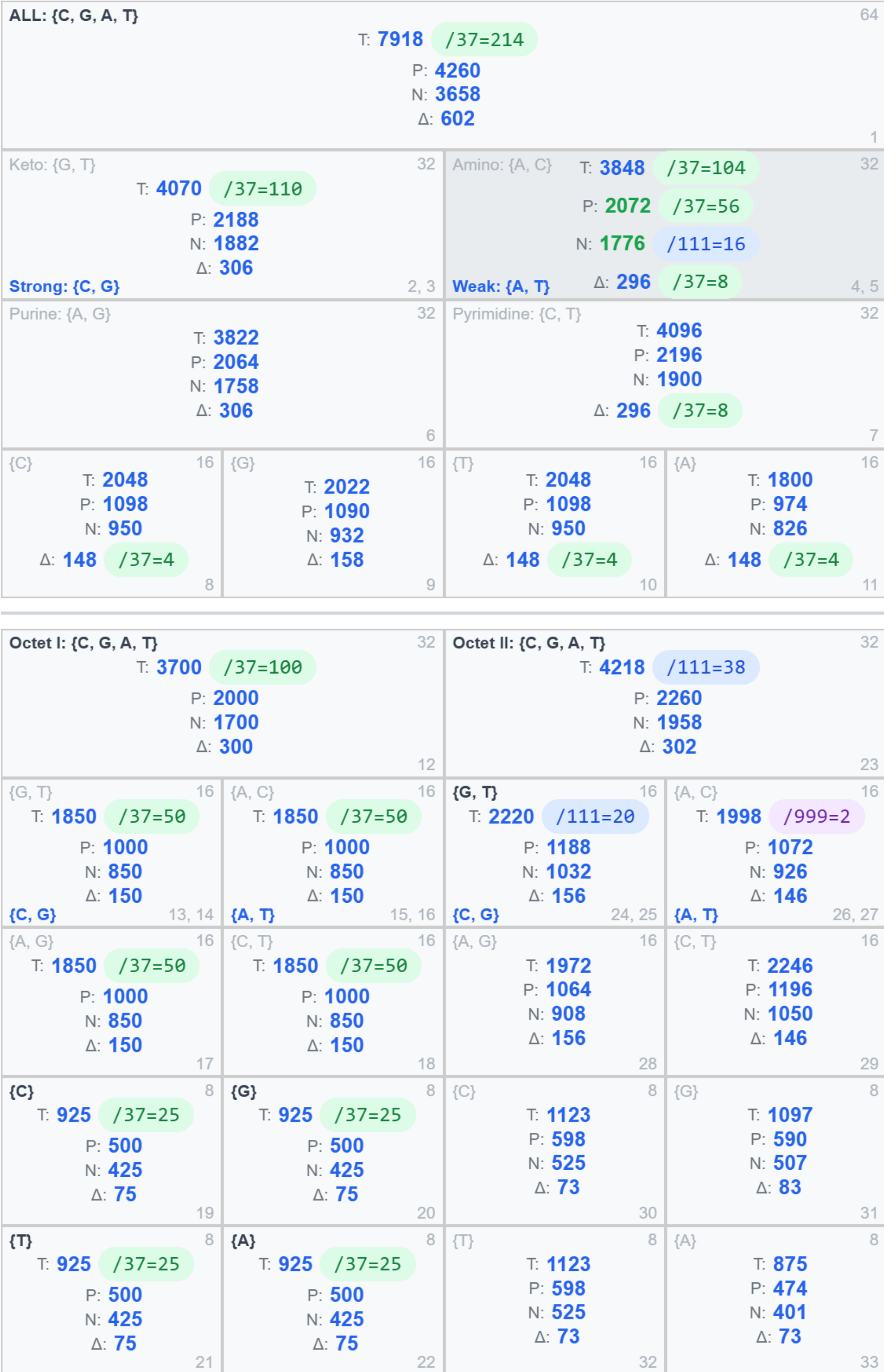

Figure 3: Nucleon parameters (T, P, N, Δ) of all codon groups under Key 0. For each cell: the group name is shown in the upper left; the group size (in codons) in the upper right; the group identifier from `codon_groups.csv` in the lower right. For twin-group pairs (Keto/Strong and Amino/Weak on the global code level, and their analogues within Octet I and Octet II), whose values are identical by pyrimidine synonymy, the display is collapsed into a single block: the second group name appears in the lower left, and the identifiers of both groups are listed, comma-separated, in the lower right. Green highlights indicate exact divisibility by 37; blue, by $111 = 3 \cdot 37$; purple, by $999 = 27 \cdot 37$. Generated by the accompanying `visualization.html`.

3.2.1 Divisibility of T by 37 at the upper levels of partitioning

Under Key 0, the total nucleon count T is divisible by 37 for the four 32-codon groups along the chemical axes Keto/Amino and Strong/Weak:

$$T(\text{Keto}) = T(\text{Strong}) = 4070 = 110 \cdot 37 \quad (\text{id 2 and id 3});$$

$$T(\text{Amino}) = T(\text{Weak}) = 3848 = 104 \cdot 37 \quad (\text{id 4 and id 5}).$$

Divisibility of T by 37 extends also to Octet II and its 16-codon sub-groups:

$$T(\text{Octet II}) = 4218 = 114 \cdot 37 \quad (\text{id 23});$$

$$T(\text{Keto} \cap \text{Octet II}) = T(\text{Strong} \cap \text{Octet II}) = 2220 = 60 \cdot 37 \quad (\text{id 24 and id 25});$$

$$T(\text{Amino} \cap \text{Octet II}) = T(\text{Weak} \cap \text{Octet II}) = 1998 = 54 \cdot 37 \quad (\text{id 26 and id 27}).$$

Combined with $T(\text{Octet I}) = 3700 = 100 \cdot 37$ from Section 3.1.2, this gives divisibility by 37 for the total nucleon count of the full code: $T(\text{ALL}) = 7918 = 214 \cdot 37$ (id 1).

The divisibilities $T(\text{Octet I}) = 3700$, $T(\text{Octet II}) = 4218$, and $T(\text{Keto}{\cap}\text{Octet II}) = T(\text{Strong}{\cap}\text{Octet II}) = 2220$ by 37 correspond, within our partitioning, to results established by Shcherbak and Makukov [8] and Panov and Filatov [5] (see Section 4.4 for further discussion); the remaining divisibilities of T by 37 given in this subsection follow algebraically from those of these groups. The numerical values of all group quantities under Key 0 are tabulated in `key0_nucleon-data.csv`.

3.2.2 Alignment of Amino and Weak with the lattice of step 37

Under Key 0, the groups Amino (id 4) and Weak (id 5) are the only 32-codon groups of the standard code in which all four nucleon quantities are simultaneously divisible by 37:

$$T = 3848 = 104 \cdot 37, \qquad P = 2072 = 56 \cdot 37,$$

$$N = 1776 = 48 \cdot 37, \qquad \Delta = 296 = 8 \cdot 37.$$

The only service codons in the Amino and Weak groups are stop codons, which contribute zero under Key 0; these sums therefore coincide with the sense-pool sums of the two groups, so the divisibility is in fact parametrization-independent, not specific to Key 0.

These same groups give anomalously simple reduced ratios:

$$\frac{P}{T} = \frac{7}{13}, \qquad \frac{N}{T} = \frac{6}{13}, \qquad \frac{\Delta}{T} = \frac{1}{13}.$$

3.2.3 The ± 1 deviation pattern under Key 0

Under Key 0 the pyrimidine Δ-cascade of Section 3.1.4 extends to further groups, and is joined by one-unit deviations of the same magnitude in the proton and neutron residues of groups in parallel positions of the codon hierarchy. The two are alike in form but distinct in the quantity affected.

The 16-codon level. The parametrization-independent condition $P \equiv +1$, $N \equiv -1 \pmod{37}$ on the 16-codon sub-groups of Octet I (Section 3.1.2) is complemented, under Key 0, by the mirror condition on the 16-codon sub-groups of the Amino branch of Octet II — $\{\text{A}, \text{C}\} \cap \text{Octet II}$ (id 26) and $\{\text{A}, \text{T}\} \cap \text{Octet II}$ (id 27):

$$P \equiv -1 \pmod{37}, \qquad N \equiv +1 \pmod{37},$$

with $P = 1072 = 29 \cdot 37 - 1$ and $N = 926 = 25 \cdot 37 + 1$.

The 8-codon level. The parametrization-independent part of the pattern at this level is described in Section 3.1.4: $\Delta = +1$ on the 8-codon sub-groups of Octet I, $\Delta = -1$ on the pyrimidine-restricted 8-codon sub-groups of Octet II. Under Key 0, $\{A\} \cap$ Octet II (id 33) joins the groups with $\Delta = 73 = 2 \cdot 37 - 1$, completing the mirror triple in Octet II. As a consequence, the parametrization-independent equality $\Delta(\{C\}) = \Delta(\{T\}) = 148$ (Section 3.1.4) extends under Key 0 to a third single-nucleotide group at the full 64-codon level: $\Delta(\{A\}) = 148$ (id 11). Under Key 0 the two stop codons of the Amino branch contribute $\Delta = 0$, so this full-code group and the 14-codon sense-pool group of the same name (Figure 2) share the value 148. The fourth single-nucleotide sub-group of Octet II, $\{G\} \cap$ Octet II (id 31), gives $\Delta = 83$ under Key 0 and is the only deviation from the pattern at this level; this group contains both the start codon ATG and the stop codon TAG.

3.3 Analytical derivation of Key 1

The parametrization-independent facts of Section 3.1, and their naive reading in Section 3.2, leave the service codons unassigned. There is a unique minimal integer assignment to those codons that extends the structure to the full 64-codon table, and this section derives it. The assignment is a device for exhibiting the arithmetic of the complete table rather than a foundation: as Section 3.2 shows, the structure is already present at the naive reading, and the derivation below only fixes the values under which it extends cleanly across the service codons.

The values are obtained from three inputs, all established earlier. The Keto/Amino asymmetry localizes in the single pair Trp–Ile, with sense-pool differences $|\delta N| = 37$ and $|\delta P| = 36$ (Section 3.1.5); the sense-pool neutron and proton totals are $97 \cdot 37$ and $\equiv -1 \pmod{37}$ (Section 3.1.1); and the proton counts are even, so $|\delta P| = 37$ is excluded by parity (Section 3.1.5). From these, Sections 3.3.1 and 3.3.2 determine the stop and start parameters through symmetry conditions—first for neutrons, then for protons—and Section 3.3.3 shows that the solution is the minimal non-negative integer element of an infinite family, and that the equality of the stop parameters and the unit offset of the start parameter follow already from two of the chemical-axis balances, without reference to the modulus. The specificity of the result across natural codes is examined in Section 4.1.

3.3.1 Determination of neutron parameters

We seek a parametrization of the service codons under which the residual deviations described in Section 3.3 disappear. We denote the (yet unknown) parameters of the stop codons by $(p_{\text{stop}}, n_{\text{stop}})$ and of the start codon by $(p_{\text{start}}, n_{\text{start}})$, with the convention that the three stop codons share the same parameters — a hypothesis already invoked in Section 3.1.5 when reducing the Keto/Amino asymmetry to the Trp–Ile pair. In this subsection we determine the neutron components n_{stop} and n_{start}. The proton components are determined in Section 3.3.2.

Distribution of service codons. All four service codons belong to Octet II; Octet I contains no service codon. Along the Keto/Amino axis, the start codon ATG and the stop codon TAG carry third-position G and belong to the Keto branch, while the stop codons TAA and TGA carry third-position A and belong to the Amino branch. Thus the Keto branch contains one stop and one start, and the Amino branch contains two stops and no start. The same distribution applies to the Strong and Weak branches respectively, by the Keto $\equiv$ Strong and Amino $\equiv$ Weak identities of Section 3.1.3.

Balance equation for Keto and Amino. The neutron pools of Keto and Amino, expressed through the sense pool values established in Section 3.1.5 and the unknown parameters of the service codons, are

$$N(\text{Keto}) = 1813 + n_{\text{start}} + n_{\text{stop}},$$
$$N(\text{Amino}) = 1776 + 2\,n_{\text{stop}}.$$

Imposing the symmetry condition $N(\text{Keto}) = N(\text{Amino})$ (the Purine/Pyrimidine axis cannot serve here, as its balance is incompatible with divisibility by 37; see Section 3.3.3) yields

$$n_{\text{stop}} - n_{\text{start}} = 37.$$

The right-hand side is the difference $N(\text{Keto sense}) - N(\text{Amino sense})$ established in Section 3.1.5; its value is determined by the nucleon difference of the Trp–Ile pair. The condition is a one-parameter family of solutions in two unknowns; a second independent constraint is required to fix the parameters individually.

Divisibility of the global neutron pool by 37. The sense pool of all 60 sense codons carries $N(\text{All sense}) = 3589 = 97 \cdot 37$ (Sections 3.1.1 and 3.1.2), an exact divisibility that is a property of the sense codons alone and independent of any service-codon parametrization. However, at the level of the entire genetic code, the inclusion of service codons disrupts this arithmetic alignment. Under Key 0, where the start codon is interpreted as Met, the global pool becomes $N(\text{All}) = 3658 = 98 \cdot 37 + 32$, which is not divisible by 37. We require the analytical parametrization to restore this underlying divisibility at the global scale:

$$N(\text{All}) = 3589 + n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37},$$

which reduces to

$$n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37}.$$

Solution. Substituting $n_{\text{stop}} = n_{\text{start}} + 37$ from the first condition into the second yields

$$4\,n_{\text{start}} + 111 \equiv 0 \pmod{37}.$$

Since $111 = 3 \cdot 37 \equiv 0 \pmod{37}$ and $\gcd(4, 37) = 1$, this reduces to

$$n_{\text{start}} \equiv 0 \pmod{37},$$

giving the family of integer solutions $n_{\text{start}} = 0, 37, 74, \ldots$ The minimal non-negative integer solution is

$$n_{\text{start}} = 0, \quad n_{\text{stop}} = 37.$$

The structure of the solution family and the minimality of this choice are addressed in Section 3.3.3.

Emergent cross-parameter equality. Substitution of the obtained values into Octet II yields

$$N(\text{Octet II}) = 1889 + 0 + 3 \cdot 37 = 2000 = P(\text{Octet I}).$$

This cross-parameter equality between the neutron content of Octet II and the proton content of Octet I was not imposed during the derivation; it emerges as a consequence of the minimal non-negative integer solution to the two conditions imposed (Keto/Amino balance and global divisibility by 37).

Non-triviality of the integer solution. Three arithmetic facts combine in the derivation, and each is independent of the other two:

1. the difference $N(\text{Keto sense}) - N(\text{Amino sense}) = 37$, a property of the Trp–Ile pair (Section 3.1.5);
2. the divisibility of the global sense pool $N(\text{All sense}) = 3589 = 97 \cdot 37$ by 37, a chemical property of the standard code independent of any service-codon parametrization;
3. the existence of a minimal non-negative integer solution with both parameters small. The relative primality $\gcd(4, 37) = 1$, where 4 is the coefficient of n_{start} in the combined equation and 37 is the modulus, ensures that the divisibility condition reduces to $n_{\text{start}} \equiv 0 \pmod{37}$ rather than to a constraint admitting multiple small residues.

Macroscopic consequences. Under the values $n_{\text{stop}} = 37$, $n_{\text{start}} = 0$, the neutron pools of the four 32-codon groups along the chemical axes satisfy

$$N(\text{Keto}) = N(\text{Amino}) = N(\text{Strong}) = N(\text{Weak}) = 1850 = \tfrac{1}{2}\,T(\text{Octet I}),$$

and the global identity

$$N(\text{All}) = T(\text{Octet I}) = 3700$$

holds. The neutron content of each chemical-axis half of the code coincides with exactly half of the T-invariant of Octet I, and the global neutron pool coincides with this T-invariant in full. The complete set of consequences is collected in Section 3.4.

3.3.2 Determination of proton parameters

Having determined the neutron components in Section 3.3.1, we now apply the same approach to determine the proton components p_{stop} and p_{start}. The convention that the three stop codons share the same parameters is preserved.

Balance equation for Keto and Amino. The proton pools of Keto and Amino, expressed through the sense pool values established in Section 3.1.5 and the unknown parameters of the service codons, are

$$P(\text{Keto}) = 2108 + p_{\text{start}} + p_{\text{stop}},$$
$$P(\text{Amino}) = 2072 + 2\,p_{\text{stop}}.$$

Imposing the corresponding condition $P(\text{Keto}) = P(\text{Amino})$ (again on the Keto/Amino axis, for the reason noted above) yields

$$p_{\text{stop}} - p_{\text{start}} = 36.$$

The right-hand side is the difference $P(\text{Keto sense}) - P(\text{Amino sense})$ established in Section 3.1.5; its value is 36 rather than 37, the parity-constrained value of $|\delta P(\text{Trp}, \text{Ile})|$ (Section 3.1.5). The condition is a one-parameter family of solutions in two unknowns; a second independent constraint is required to fix the parameters individually.

Divisibility of the global proton pool by 37. The sense pool of all 60 sense codons of the standard code carries

$$P(\text{All sense}) = P(\text{Octet I}) + P(\text{Octet II sense}) = 2000 + 2180 = 4180 = 113 \cdot 37 - 1.$$

Unlike the neutron pool, the global proton pool of the sense codons is not divisible by 37; it lies one proton below the nearest multiple, $P(\text{All sense}) \equiv -1 \pmod{37}$. We require the analytical parametrization of the service codons to close this deficit and bring the global proton pool onto the lattice of step 37:

$$P(\text{All}) = 4180 + p_{\text{start}} + 3\, p_{\text{stop}} \equiv 0 \pmod{37},$$

which reduces to

$$p_{\text{start}} + 3\, p_{\text{stop}} \equiv 1 \pmod{37}.$$

Solution. Substituting $p_{\text{stop}} = p_{\text{start}} + 36$ from the first condition into the second yields

$$4\, p_{\text{start}} + 108 \equiv 1 \pmod{37}.$$

Since $108 = 2 \cdot 37 + 34$ and $\gcd(4, 37) = 1$, this reduces to

$$4\, p_{\text{start}} \equiv 4 \pmod{37}, \quad \text{hence} \quad p_{\text{start}} \equiv 1 \pmod{37},$$

giving the family of integer solutions $p_{\text{start}} = 1, 38, 75, \ldots$ The minimal non-negative integer solution is

$$p_{\text{start}} = 1, \quad p_{\text{stop}} = 37.$$

The structure of the solution family and the minimality of this choice are addressed in Section 3.3.3.

Emergent equality of stop parameters. The neutron and proton parameters of the stop codons obtained independently in Sections 3.3.1 and 3.3.2 coincide:

$$p_{\text{stop}} = n_{\text{stop}} = 37.$$

Each stop codon under Key 1 carries the same number of protons and neutrons, both equal to the organizing multiplier 37. This equality was not imposed; it follows from the minimal non-negative integer solutions to two formally independent systems — one in neutrons, one in protons — each constrained by Keto/Amino balance and global divisibility by 37.

Non-triviality of the integer solution. Three arithmetic facts combine in the derivation:

1. the difference $P(\text{Keto sense}) - P(\text{Amino sense}) = 36$, the parity-constrained value of $|\delta P(\text{Trp}, \text{Ile})|$ (Section 3.1.5);

2. the residue $P(\text{All sense}) \equiv -1 \pmod{37}$, a deficit of one proton on the 37-lattice and a chemical property of the standard code independent of any service-codon parametrization;

3. the existence of a minimal non-negative integer solution with both parameters small. The relative primality $\gcd(4, 37) = 1$, where 4 is the coefficient of p_{start} in the combined equation and 37 is the modulus, ensures that the divisibility condition reduces to $p_{\text{start}} \equiv 1 \pmod{37}$ rather than to a constraint admitting multiple small residues.

Facts (1) and (2) share a common origin in the proton parity of the canonical amino acids: parity excludes odd values on the 37-lattice — both the value 37 for $|\delta P|$ and an odd multiple for $P(\text{All sense})$ — forcing an unavoidable deviation from these target values. The particular values that this deviation takes (36 and -1) are fixed by the nucleon composition of the code. The minimal solution $p_{\text{start}} = 1$ closes exactly this single-proton deficit at the global scale.

Macroscopic consequences. Under the values $p_{\text{stop}} = 37$, $p_{\text{start}} = 1$, the proton pools of the four 32-codon groups along the chemical axes satisfy

$$P(\text{Keto}) = P(\text{Amino}) = P(\text{Strong}) = P(\text{Weak}) = 2146 = \tfrac{1}{2}\,T(\text{Octet II}),$$

and the global identity

$$P(\text{All}) = T(\text{Octet II}) = 4292$$

holds, with $T(\text{Octet II})$ evaluated under Key 1. Unlike $T(\text{Octet I}) = 3700$, which is parametrization-independent and serves as a chemical invariant, $T(\text{Octet II})$ depends on the values assigned to the service codons; under Key 1 it takes the value 4292. Together with the analogous neutron identities $N(\text{All}) = T(\text{Octet I}) = 3700$ and $N(\text{Keto}) = \frac{1}{2}\,T(\text{Octet I})$ established in Section 3.3.1, the parametrization realizes a dual matching: the global neutron pool aligns with $T(\text{Octet I})$, the global proton pool aligns with $T(\text{Octet II})$, and each chemical-axis half of the code coincides with exactly half of the corresponding octet's total nucleon content. The complete set of consequences is collected in Section 3.4.

3.3.3 Hierarchy of constraints and the choice of Key 1

The derivation in Sections 3.3.1 and 3.3.2 imposed two kinds of constraint on the service-codon parameters, each applied to neutrons and to protons: Keto/Amino balance and divisibility of the global pool by 37. Each constraint reduces the dimension of the solution space, and the family of admissible parametrizations narrows in three steps:

1. Keto/Amino balance alone, applied separately to neutrons and protons, gives a two-parameter family of solutions: the conditions $n_{\text{stop}} - n_{\text{start}} = 37$ and $p_{\text{stop}} - p_{\text{start}} = 36$ reduce the four service-codon parameters to two independent degrees of freedom, one for each nucleon quantity. All elements of this family align the four 32-codon groups along the chemical axes Keto / Amino / Strong / Weak.

2. Adding global divisibility of $N(\text{All})$ by 37 further restricts the neutron parameters to the discrete one-parameter family $n_{\text{start}} \in \{0, 37, 74, \ldots\}$, that is $n_{\text{start}} = 37k$ for $k \in \mathbb{Z}_{\geq 0}$. Adding global divisibility of $P(\text{All})$ by 37 similarly restricts the proton parameters to $p_{\text{start}} \in \{1, 38, 75, \ldots\}$, that is $p_{\text{start}} = 1 + 37m$ for $m \in \mathbb{Z}_{\geq 0}$. The combined family becomes a discrete two-dimensional lattice indexed by $(k, m) \in \mathbb{Z}^2_{\geq 0}$ with step 37 in each direction.

3. Selecting the minimal non-negative integer element of this lattice ($k = m = 0$) gives Key 1: $(p_{\text{stop}}, n_{\text{stop}}) = (37, 37)$, $(p_{\text{start}}, n_{\text{start}}) = (1, 0)$.

The minimal element is distinguished from all other solutions by a combination of structural features that the constraints alone do not require: the value $p_{\text{stop}} = n_{\text{stop}} = 37$, equal to the organizing multiplier of the code and attained only at the minimal element; the value $p_{\text{start}} = 1$, closing exactly the parity-induced deficit $P(\text{All sense}) \equiv -1 \pmod{37}$ (Section 3.3.2); and the value $n_{\text{start}} = 0$, at which the start codon contributes no neutrons. Of these, only $p_{\text{stop}} = n_{\text{stop}} = 37$ singles out the minimal element on its own: $n_{\text{start}} = 0$ holds for every element with $k = 0$ and $p_{\text{start}} = 1$ for every element with $m = 0$, and the two coincide with $p_{\text{stop}} = n_{\text{stop}} = 37$ only at $k = m = 0$. Higher elements of the lattice satisfy the same constraints but lack this combination. We adopt Key 1 as the parametrization of the standard genetic code throughout the remainder of the paper.

The stop-parameter equality as a two-axis condition. The equality $p_{\text{stop}} = n_{\text{stop}}$ and the offset $p_{\text{start}} - n_{\text{start}} = 1$, obtained in Section 3.3.3 from the minimal solution of the two modulo-37 systems, also follow from the two chemical-axis balances alone, with no reference to 37. Writing $\Delta = P - N$, the Keto/Amino balance ($n_{\text{stop}} - n_{\text{start}} = 37$, $p_{\text{stop}} - p_{\text{start}} = 36$) gives

$\Delta_{\text{start}} = \Delta_{\text{stop}} + 1$. Since all four service codons lie in the Purine branch, equalizing the Δ-invariant across the Purine/Pyrimidine axis requires the service increment to close the sense-pool deficit $\Delta(\text{Pyr sense}) - \Delta(\text{Pur sense}) = 1$, that is $\Delta_{\text{start}} + 3\,\Delta_{\text{stop}} = 1$. The same purine localization of the service codons is also what forces the balance in P and N to be imposed on the Keto/Amino axis rather than on Purine/Pyrimidine: with all increments confined to the Purine branch, a Purine = Pyrimidine balance would require $p_{\text{start}} + 3\,p_{\text{stop}} = 212$ and $n_{\text{start}} + 3\,n_{\text{stop}} = 211$, whereas global divisibility requires these same sums to be $\equiv 1$ and $\equiv 0 \pmod{37}$; since $212 \equiv 27$ and $211 \equiv 26$, the two demands are incompatible, and among the third-position axes only a balance on Keto/Amino (equivalently, by the twin identity of Section 3.1.3, on Strong/Weak) is compatible with the 37-lattice. The obstruction is specific to P and N, whose service sums are large (212 and 211) and land at the wrong residues. For $\Delta = P - N$ the situation differs: the Purine/Pyrimidine Δ-deficit is only $\Delta(\text{Pyr sense}) - \Delta(\text{Pur sense}) = 1$, small enough to be closed by the service increment, and it coincides with the offset $\Delta_{\text{start}} - \Delta_{\text{stop}} = 1$ left by the Keto/Amino balance. The Keto/Amino P/N balance and the Purine/Pyrimidine Δ-balance are therefore mutually compatible, which is what makes the two-axis condition available.

This fixes only the proton–neutron differences of the service codons, not their absolute values: the modulus 37 and the minimality condition remain necessary to obtain $(37, 37)$ and $(1, 0)$. Its content is that the equality of the stop parameters is not tied to the modulus—it holds whenever the Keto/Amino and Purine/Pyrimidine Δ-balances hold—and that it reduces to the agreement of two sense-pool unit offsets: the proton–neutron difference $\delta P - \delta N = -1$ of the localizing pair (Section 3.1.5) and the branch deficit $\Delta(\text{Pyr sense}) - \Delta(\text{Pur sense}) = +1$. The coincidence is thus relocated rather than removed, but it is shown to be a joint property of two chemical axes rather than an artifact of the parametrization.

Convergence of independent routes. The values of Key 1 are not tied to a single derivation. Three routes, resting on different structural inputs, reach the same assignment. The first is the modulo-37 system used above: the Keto/Amino balance together with divisibility of the global neutron and proton pools, whose minimal nonnegative solution is $(p_{\text{stop}}, n_{\text{stop}}) = (37, 37)$, $(p_{\text{start}}, n_{\text{start}}) = (1, 0)$. The second is the two-axis condition of Section 3.3.3: the Keto/Amino balance together with the Δ-balance across the Purine/Pyrimidine axis forces $p_{\text{stop}} = n_{\text{stop}}$ and $p_{\text{start}} = n_{\text{start}} + 1$ with no reference to the modulus. The third fixes the neutron parameters directly from the parametrization-independent half-octet target $N(\text{Amino}) = N(\text{Keto}) = \frac{1}{2}\,T(\text{Octet I}) = 1850$: the sense-pool deficits $1850 - 1776 = 74 = 2 \cdot 37$ (closed by the two stop codons of the Amino branch) and $1850 - 1813 = 37$ (closed by the stop and start of the Keto branch) give $n_{\text{stop}} = 37$ and $n_{\text{start}} = 0$.

3.4 Consequences under Key 1

In this section we describe the arithmetic structure of the standard genetic code under Key 1: alignment of the 32-codon chemical-axis groups, identities between the global nucleon pools and the structural totals of the octets, higher-order divisibility properties, the Δ-invariant across all six 32-codon groups, and the Purine/Pyrimidine asymmetry. Figure 4 shows the same overview computed under Key 1. The four chemical-axis groups Keto, Strong, Amino, and Weak now become identical in T, P, N, and Δ, accounting for the visible uniformity in the top half of the figure.

3.4.1 Alignment of the chemical-axis 32-codon groups

Under Key 1, the four 32-codon groups along the chemical axes Keto/Amino and Strong/Weak are aligned in all four nucleon quantities to the common values

$$T = 3996, \quad P = 2146, \quad N = 1850, \quad \Delta = 296.$$

Figure 4: The same overview as Figure 3, computed under the analytically derived parametrization Key 1, in which all four service codons are treated as formal parameters: the three stop codons contribute $(P, N) = (37, 37)$ each, and ATG contributes $(P, N) = (1, 0)$ (the methionine value of ATG is not used). Under this parametrization the four chemical-axis 32-codon groups Keto, Strong, Amino, Weak become identical in T, P, N, and Δ (top half of the figure). Generated by the accompanying `visualization.html`.

The equality Keto = Amino follows from the balance conditions imposed during the derivation of n_{stop} and p_{stop}; the identities Keto ≡ Strong and Amino ≡ Weak are parametrization-independent (Section 3.1.3). The numerical values of all group quantities under Key 1 are tabulated in `key1_nucleon-data.csv`.

Of the three axes that partition the codon table by third position — Keto/Amino, Strong/Weak, and Purine/Pyrimidine — only the first two yield this alignment. The Purine/Pyrimidine axis behaves differently and is analyzed in Section 3.4.5.

Rational signature. The alignment can be restated in projective form. Under Key 1, the whole code and the four chemical-axis 32-codon groups share a single rational profile of nucleon ratios,

$$\frac{P}{T} = \frac{29}{54}, \qquad \frac{N}{T} = \frac{25}{54}, \qquad \frac{\Delta}{T} = \frac{2}{27},$$

with $P : N = 29 : 25$. The two terms of this ratio are $T(L_{16} \cap \text{Octet II})/(2 \cdot 37) = 29$ and $T(L_{16} \cap \text{Octet I})/(2 \cdot 37) = 25$, so the profile carries the same pairing of P with Octet II and N with Octet I as the macroscopic identities of Section 3.4.2. The ratio profiles of all groups are tabulated in `key1_ratios.csv`.

3.4.2 Macroscopic identities and propagation to chemical-axis halves

Under Key 1, the global nucleon pools of the standard code satisfy three identities relating them to the structural totals of the two octets:

$$\begin{aligned} N(\text{All}) &= T(\text{Octet I}) = 3700, \\ P(\text{All}) &= T(\text{Octet II}) = 4292, \\ N(\text{Octet II}) &= P(\text{Octet I}) = 2000. \end{aligned}$$

The first identity records the cross-parameter equality already noted in Section 3.3.1: the global neutron pool of the code under Key 1 equals the parametrization-independent chemical constant $T(\text{Octet I})$. The second is the analogue on the proton side, with $P(\text{All}) = 4292$ matching $T(\text{Octet II})$ evaluated under Key 1; unlike $T(\text{Octet I})$, the right-hand side here depends on the service-codon parametrization. The third identity restates the first at the level of individual octets: since $T(\text{Octet I}) = P(\text{Octet I}) + N(\text{Octet I})$ tautologically, with chemical constants $P(\text{Octet I}) = 2000$ and $N(\text{Octet I}) = 1700$, the equality $N(\text{All}) = T(\text{Octet I})$ from the first identity gives $N(\text{Octet II}) = 2000 = P(\text{Octet I})$.

Combining the macroscopic identities with the alignment of the four chemical-axis groups (Section 3.4.1) gives

$$N(\text{Keto}) = N(\text{Amino}) = N(\text{Strong}) = N(\text{Weak}) = 1850 = \tfrac{1}{2}\,T(\text{Octet I}),$$

$$P(\text{Keto}) = P(\text{Amino}) = P(\text{Strong}) = P(\text{Weak}) = 2146 = \tfrac{1}{2}\,T(\text{Octet II}).$$

Each chemical-axis half of the code carries exactly half of the structural total of the corresponding octet. The same two values reappear at the L_{16} level inside the octets. Each of the six L_{16} sub-groups of Octet I carries $T = 1850$, equal across the six sub-groups because each 4-fold degenerate family of Octet I contributes the same amino acid nucleon to every L_{16} partition. Inside Octet II, the four L_{16} sub-groups along the chemical axes Keto/Amino and Strong/Weak carry $T = 2146$ under Key 1; the two L_{16} sub-groups along the Purine/Pyrimidine axis are not aligned with this value (Section 3.4.5). The cross-parameter equality $N(\text{Octet II}) = P(\text{Octet I}) = 2000$ similarly resolves at the L_{16} level: each L_{16} sub-group of Octet I carries $P = 1000$, and each chemical-axis L_{16} sub-group of Octet II carries $N = 1000$ under Key 1. The full set of equalities arising under Key 1, classified by relation type, is tabulated in `key1_equalities.csv`.

A parametrization-independent counterpart on the proton side. The value 2000 that appears in the cross-parameter identity $N(\text{Octet II}) = P(\text{Octet I}) = 2000$ also arises elsewhere in the arithmetic of the code, on the proton side and without reference to any service-codon parametrization. The sense pool of Octet II carries $P(\text{Octet II sense}) = 2180$; subtracting the single pair Trp and Ile that localizes the Keto/Amino asymmetry (Section 3.1.5) gives

$$P(\text{Octet II sense}) - \big(P(\text{Trp}) + P(\text{Ile})\big) = 2180 - 180 = 2000 = P(\text{Octet I}).$$

The proton excess of Octet II over Octet I at the sense-pool level coincides exactly with the proton count of the localizing pair: $P(\text{Trp})+P(\text{Ile}) = 180 = P(\text{Octet II sense})-P(\text{Octet I})$. The analogous identity for neutrons does not hold — $N(\text{Trp})+N(\text{Ile}) = 155$, while $N(\text{Octet II sense})-N(\text{Octet I}) = 189$, a residual difference of 34 — and the failure of the neutron analogue indicates that the proton equality is a numerical coincidence rather than a structural regularity. It is recorded as an illustration of the recurrence of the value 2000 and carries no independent evidential weight.

3.4.3 Divisibility by 37 and by 999

The arithmetic of Key 1 is organized by two divisibility properties of the standard code. The divisibility by 37 was imposed during the derivation of n_{stop} and p_{stop} for the global pools $N(\text{All})$ and $P(\text{All})$; under Key 1 it extends to many further group quantities. The divisibility by 999 is not required by the system and holds independently for the total nucleon pool.

Divisibility by 37. Under Key 1, divisibility by 37 holds for the global neutron and proton pools, and for the 32-codon groups along the chemical axes Keto/Amino and Strong/Weak in T, P, N, and Δ:

$$\begin{aligned}
N(\text{All}) &= 100 \cdot 37,\\
P(\text{All}) &= 116 \cdot 37,\\
T(\text{Keto}) = T(\text{Amino}) = T(\text{Strong}) = T(\text{Weak}) &= 108 \cdot 37,\\
P(\text{Keto}) = P(\text{Amino}) = P(\text{Strong}) = P(\text{Weak}) &= 58 \cdot 37,\\
N(\text{Keto}) = N(\text{Amino}) = N(\text{Strong}) = N(\text{Weak}) &= 50 \cdot 37,\\
\Delta(\text{Keto}) = \Delta(\text{Amino}) = \Delta(\text{Strong}) = \Delta(\text{Weak}) &= 8 \cdot 37.
\end{aligned}$$

By the first two macroscopic identities of Section 3.4.2, divisibility by 37 also extends to the global proton-neutron imbalance: $\Delta(\text{All}) = P(\text{All}) - N(\text{All}) = T(\text{Octet II}) - T(\text{Octet I}) = 16 \cdot 37 = 592$. The full set of divisibility records under Key 1 is tabulated in `key1_divisibility-37.csv`.

Divisibility by 999. Under Key 1, the global total nucleon pool of the standard code and each of the four chemical-axis 32-codon groups in T satisfy

$$T(\text{All}) = 7992 = 8 \cdot 999, \quad T(\text{Keto}) = T(\text{Amino}) = T(\text{Strong}) = T(\text{Weak}) = 3996 = 4 \cdot 999.$$

The system of conditions imposed in Section 3.3 requires divisibility by 37 but not by the higher modulus $999 = 27 \cdot 37$. In factored form $T(\text{All}) = 2^3 \cdot 3^3 \cdot 37 = 6^3 \cdot 37$. The divisibility of $T(\text{All})$ and $T(\text{half})$ by 999 is a further property of the standard code under Key 1; being neither required by the derivation nor parametrization-independent, it is recorded but carries no independent evidential weight.

3.4.4 The Δ-invariant across all six 32-codon groups

Under Key 1, all six 32-codon groups defined by the third codon position carry the same value of the proton-neutron difference:

$$\Delta(\text{Keto}) = \Delta(\text{Amino}) = \Delta(\text{Strong}) = \Delta(\text{Weak}) = \Delta(\text{Purine}) = \Delta(\text{Pyrimidine}) = 296.$$

For the four groups along the chemical axes Keto/Amino and Strong/Weak, this follows from the alignment in P and N established in Section 3.4.1: equal P and equal N imply equal Δ. For the Purine and Pyrimidine groups, which are not aligned in T, P, or N (Section 3.4.5), the Δ-equality is independent of the alignment: the P-shift and the N-shift between the two groups coincide in magnitude, so that their difference $\Delta = P - N$ is preserved across all three axes.

The common value $\Delta = 296$ resolves through a cascade joining the lower partition levels. At the 8-codon level, every $L_8 \cap \text{Octet I}$ carries $\Delta = 75$ as a parametrization-independent property of the 4-fold-degenerate families (Section 3.1.2), and every $L_8 \cap \text{Octet II}$ carries $\Delta = 73$ under Key 1 (paralleling the Δ-pattern of Section 3.1.4):

$$\Delta(L_8 \cap \text{Octet I}) = 75, \qquad \Delta(L_8 \cap \text{Octet II}) = 73.$$

The 16-codon level reproduces these values along two distinct partitions of the code. Within each octet, every L_{16} sub-group inherits twice the corresponding L_8 value,

$$\Delta(L_{16} \cap \text{Octet I}) = 150, \qquad \Delta(L_{16} \cap \text{Octet II}) = 146.$$

Across octets, each single-base 16-codon group of the full code joins one $L_8 \cap \text{Octet I}$ with one $L_8 \cap \text{Octet II}$, giving

$$\Delta(\{\text{C}\}) = \Delta(\{\text{G}\}) = \Delta(\{\text{T}\}) = \Delta(\{\text{A}\}) = 75 + 73 = 148 = 4 \cdot 37$$

at All-code level (ids 8–11). This four-way equality is specific to Key 1: under Key 0 the placement of the start codon ATG as methionine in {G} shifts $\Delta(\{\text{G}\})$ to 158, breaking the symmetry with the remaining three single-base groups. The two 16-codon paths converge at the 32-codon level: the common value $\Delta = 296 = 2 \cdot 148 = 150 + 146$ obtains either as the sum of two single-base 16-codon groups along any axis, or equivalently as the sum of an $L_{16} \cap \text{Octet I}$ and an $L_{16} \cap \text{Octet II}$ from the same axis.

3.4.5 Purine/Pyrimidine asymmetry

The Purine/Pyrimidine axis is the one chemical axis on which the standard code is not aligned in T, P, or N. Under Key 1,

$$\begin{aligned} N(\text{Pyrimidine}) - N(\text{Purine}) &= 100, \\ P(\text{Pyrimidine}) - P(\text{Purine}) &= 100, \\ T(\text{Pyrimidine}) - T(\text{Purine}) &= 200, \\ \Delta(\text{Pyrimidine}) - \Delta(\text{Purine}) &= 0. \end{aligned}$$

The asymmetry is localized in Octet II. Within Octet I, Purine and Pyrimidine contain equal proton and neutron pools as a parametrization-independent consequence of 4-fold degeneracy (Section 3.1); the imbalance arises entirely from Octet II, where all four service codons reside in the Purine branch (third position A or G).

The shifts in P and N are equal in magnitude (100 in each), and this magnitude equality is what preserves the Δ-invariant across the Purine and Pyrimidine groups (Section 3.4.4). At the level of the sense pool alone, the two shifts are not equal:

$$P(\text{Pyr sense}) - P(\text{Pur sense}) = 212, \quad N(\text{Pyr sense}) - N(\text{Pur sense}) = 211,$$

with a residual proton-neutron asymmetry of one in favor of P. Under Key 1, the service codons add 112 protons and 111 neutrons to the Purine branch: one ATG with $p_{\text{start}} = 1$, $n_{\text{start}} = 0$, plus three stop codons each with $p_{\text{stop}} = n_{\text{stop}} = 37$. The proton-neutron asymmetry of the service increments matches that of the sense pool. Adding these increments to the Purine branch reduces the Pyr–Pur proton difference from 212 to 100 and the neutron difference from 211 to 100, equalizing the two macroscopic shifts.

Purine synonymy at the single-base level under Key 1. At the 16-codon level, the four single-base groups of the full code split into two pairs by their (T, P, N) profiles:

$$T(\{\mathrm{C}\}) = T(\{\mathrm{T}\}) = 2048, \quad P(\{\mathrm{C}\}) = P(\{\mathrm{T}\}) = 1098, \quad N(\{\mathrm{C}\}) = N(\{\mathrm{T}\}) = 950,$$
$$T(\{\mathrm{G}\}) = T(\{\mathrm{A}\}) = 1948, \quad P(\{\mathrm{G}\}) = P(\{\mathrm{A}\}) = 1048, \quad N(\{\mathrm{G}\}) = N(\{\mathrm{A}\}) = 900,$$

while Δ remains uniform across all four groups (Section 3.4.4). The pyrimidine equality $\{\mathrm{C}\} \equiv \{\mathrm{T}\}$ is parametrization-independent and follows from the universal C/T synonymy at the family level (Section 3.1.3). The purine equality $\{\mathrm{G}\} \equiv \{\mathrm{A}\}$ is specific to Key 1, with the mechanism local to Octet II: the branches $\{\mathrm{G}\} \cap$ Octet II and $\{\mathrm{A}\} \cap$ Octet II contain asymmetrically distributed service codons (ATG and TAG in $\{\mathrm{G}\}$ versus TAA and TGA in $\{\mathrm{A}\}$), and the symmetrized Key 1 values cancel the corresponding sense-pool asymmetries exactly:

$$\{\mathrm{G}\} \cap \text{Octet II} = \{\mathrm{A}\} \cap \text{Octet II} = (T = 1023,\ P = 548,\ N = 475).$$

The Pyrimidine pair and the Purine pair differ by exactly 100 in T and 50 in each of P and N; doubling these shifts reproduces the macroscopic Purine/Pyrimidine asymmetry at the 32-codon level.

Inside Octet II at the 16-codon level. The L_{16} sub-groups along the Purine/Pyrimidine axis carry

$$N(\text{Pur} \cap \text{Octet II}) = 950, \quad N(\text{Pyr} \cap \text{Octet II}) = 1050,$$
$$P(\text{Pur} \cap \text{Octet II}) = 1096, \quad P(\text{Pyr} \cap \text{Octet II}) = 1196,$$

showing a ± 50 deviation from the values carried by the L_{16} sub-groups of Octet II along the Keto/Amino and Strong/Weak axes, which under Key 1 are $N = 1000$ and $P = 1146$ (Section 3.4.2). The Purine and Pyrimidine branches of Octet II thus straddle the values of the other two axes symmetrically.

3.5 Synthesis

The results above separate along the three levels of Section 3. The parametrization-independent level (Section 3.1) carries the structure: the sense-pool neutron divisibility and proton residue, the localization of the Keto/Amino asymmetry in a single amino-acid pair, and the Octet I and pyrimidine constants. These are the load-bearing findings, and they need no assignment to the service codons. The naive reading (Section 3.2) already exhibits much of the same structure. Key 1 (Sections 3.3–3.4) adds no new structure: it is the unique minimal integer assignment under which the parametrization-independent regularities extend cleanly to the full 64-codon table. Within the Key 1 level a distinction remains between relations that follow algebraically from the imposed balance conditions and relations that hold under Key 1 without following from them; this subsection collects the results under these headings.

Why the extension is solvable in small integers. The parametrization-independent facts of Section 3.1 are what make the service-codon system of Sections 3.3.1–3.3.2 solvable in small non-negative integers. Three enter directly: the Keto/Amino asymmetry reduces to the single pair Trp–Ile with $|\delta N| = 37$ and $|\delta P| = 36$ (Section 3.1.5); the sense pool carries $N(\text{All sense}) = 97 \cdot 37$ and $P(\text{All sense}) \equiv -1 \pmod{37}$ (Section 3.1.1); and the even-proton property forbids $|\delta P| = 37$ (Section 3.1.5). The deficits then close arithmetically: the neutron deficit $T(\text{Octet I}) - N(\text{All sense}) = 3 \cdot 37 = 111$ is a multiple both of the number of stops and of the modulus 37; the proton residue -1 is the smallest non-zero contribution the start codon can make; and $\gcd(4, 37) = 1$ reduces the combined condition to a unique minimal solution rather than one admitting multiple small residues. The neutron and proton systems are formally independent, yet their minimal solutions coincide at $p_{\text{stop}} = n_{\text{stop}} = 37$. Across the alternative NCBI tables (Section 4.1) this arithmetic match holds only for the standard code and its exact codon-to-amino-acid equivalent.

Algebraic consequences of the Key 1 system. The balance conditions imposed during the derivation directly equate the Keto and Amino groups in N and P. Combined with the parametrization-independent identities Keto $\equiv$ Strong and Amino $\equiv$ Weak (Section 3.1.3), this aligns the four 32-codon groups along the Keto/Amino and Strong/Weak axes in all four nucleon quantities (T, P, N, Δ) (Section 3.4.1). From the alignment follow the half-code identities $N(\text{half}) = \frac{1}{2}T(\text{Octet I})$ and $P(\text{half}) = \frac{1}{2}T(\text{Octet II})$, and the divisibility of their nucleon quantities by 37.

Independent observations under Key 1. The remaining identities hold under the parametrization without following from the balance conditions. The global neutron pool reaches $N(\text{All}) = 3700$, the minimal admissible multiple of 37, which coincides with the parametrization-independent constant $T(\text{Octet I})$; the equivalent statement at the level of individual octets is $N(\text{Octet II}) = P(\text{Octet I}) = 2000$, a cross-parameter equality between a parametric quantity and a chemical constant. The analogous proton identity $P(\text{All}) = T(\text{Octet II}) = 4292$ holds under the same parametrization, completing the dual pairing between the global nucleon pools and the two octets, and follows from the neutron identity together with $T = P + N$ (Section 3.4.2). The total nucleon pool satisfies $T(\text{All}) = 8 \cdot 999$, a divisibility by the higher modulus 999 not present among the imposed conditions (Section 3.4.3). The Δ-invariant $\Delta = 296$ extends to all six 32-codon groups, including the Purine and Pyrimidine groups, which are not aligned in T, P or N (Section 3.4.4). On the Purine/Pyrimidine axis, the proton and neutron shifts between the two groups are equal and both take the value 100, with the equality of the two shifts following from the matching of the parity-induced unit shift between the sense pool and the parametric increments (Section 3.4.5). At the single-base level, the purine equality $\{\text{G}\} \equiv \{\text{A}\}$ holds under Key 1 in addition to the parametrization-independent pyrimidine equality $\{\text{C}\} \equiv \{\text{T}\}$. A parametrization-independent counterpart on the proton side gives $P(\text{Octet II sense}) - \big(P(\text{Trp}) + P(\text{Ile})\big) = P(\text{Octet I}) = 2000$: the proton excess of Octet II over Octet I on the sense pool coincides exactly with the proton mass of the localizing pair, an identity that does not hold for neutrons.

A proton-side offset of one unit. In the parametrization-independent arithmetic, two neutron quantities meet the 37-lattice exactly while the corresponding proton quantities fall one unit short. The sense pool carries $N(\text{All sense}) = 97 \cdot 37$ against $P(\text{All sense}) = 113 \cdot 37 - 1$; the single pair Trp–Ile that localizes the Keto/Amino asymmetry has $|\delta N| = 37$ against $|\delta P| = 36$. The derived parametrization then closes the offset with one extra proton on the start codon, $p_{\text{start}} = 1$, where the neutron side requires none, $n_{\text{start}} = 0$. The common cause is the even-proton property of the canonical amino acids (Section 3.1.5): a sum or difference of even proton counts is even and cannot equal an odd multiple of 37, so both 37 itself and $113 \cdot 37$ are unreachable on the proton side. That the miss is exactly one unit in each case, rather than a larger margin, is a property of the specific nucleon counts and is not fixed by parity alone.

Two recurrent integers: 37 and the round-number scale. The integer 37 appears at all three levels of the structure and in roles not directly connected by the derivation. At the parametrization-independent level it is the common denominator of the Octet I ratios, the divisor of the sense pool $N(\text{All sense}) = 97 \cdot 37$, and the value of $|\delta N(\text{Trp}, \text{Ile})|$. Under Key 0 it organizes the alignment of the Amino and Weak groups and their sub-group $\{A\}$ with the 37-lattice (Section 3.2). Under Key 1 it is the modulus of the imposed divisibility conditions, the common value of the stop-codon parameters, and a divisor of the nucleon quantities of the aligned 32-codon groups. These occurrences meet at the cross-parameter identity $N(\text{All}) = T(\text{Octet I})$, where the parametrization-independent value $100 \cdot 37$ and the Key 1-level divisibility of $N(\text{All})$ by 37 fall on the same number. That a topological quantity of Octet I, a global neutron sum, and the neutron difference of a single amino-acid pair are all expressed through the same modulus, although none of these is derived from the others, indicates that the regularities form a connected whole rather than separate coincidences; the identities relating them, such as $T(\text{Octet I}) - N(\text{All sense}) = 3 \cdot |\delta N(\text{Trp}, \text{Ile})|$, follow once the shared modulus is granted and add no independent divisibility of their own.

A second integer scale is also visible in the arithmetic of the code. The parametrization-independent constants of Octet I — $P(\text{Octet I}) = 2000$, $N(\text{Octet I}) = 1700$, $\Delta(\text{Octet I}) = 300$ — are all multiples of 100, with $P(\text{Octet I})$ a multiple of 1000. Under Key 1 the cross-parameter equality $N(\text{Octet II}) = P(\text{Octet I}) = 2000$ brings the neutron content of Octet II onto the same scale, and the neutron pools of the Purine and Pyrimidine groups satisfy $N(\text{Purine}) = 1800$ and $N(\text{Pyrimidine}) = 1900$. The Purine/Pyrimidine shifts in P and N take the value 100 at the 32-codon level and ± 50 at the L_{16} level. These round-number occurrences do not reduce to the multiplier 37 and constitute a separate arithmetic regularity of the standard code under Key 1.

4 Discussion

This section examines how far the regularities reported above are specific to the standard code and its structure, rather than artifacts of the analysis. Section 4.1 compares the standard code against the alternative NCBI codes on the sense-pool arithmetic, testing both the global alignment with the 37-lattice and the robustness of the Trp–Ile localization; Section 4.2 asks whether the third codon position and the modulus 37 are distinguished from other positions and divisors; and Section 4.3 assesses the concentration on the 37-lattice against a random-code null. Section 4.4 situates the results relative to prior work, and Section 4.5 states the limitations and outlook.

4.1 Specificity across alternative genetic codes

The standard genetic code is one of 27 translation tables in the NCBI registry. The parametrization-independent sense-pool arithmetic of Section 3.1 admits direct comparative tests across these tables: for each alternative code one can ask whether the same arithmetic holds. Two such tests are reported here. The first concerns the global alignment of the sense pool with the lattice of step 37—the neutron deficit and the proton residue of Section 3.1.1. The second concerns the localization of the Keto/Amino asymmetry in the single pair Trp–Ile (Section 3.1.5). Both quantities are properties of the sense pool and require no assignment to the service codons; the closure of the first pair under Key 1 is a separate matter, treated in Section 3.3. The two tests probe different aspects of the arithmetic and, as shown below, identify different sets of matching alternative codes.

Because both tests depend on which codons are counted as stops, both are computed under two model choices for the sense pool:

- Sense pool, Model 1 (primary): all codons listed as stops in the NCBI registry are excluded, even when they are read as amino acids in context-dependent codes. The primary start codon ATG is excluded as the universal initiator. Initiation is a property of the first codon position rather than of the triplet itself, so exactly one initiator codon is removed; alternative start codons such as GTG and TTG carry their elongation meaning at every internal position and are counted as sense codons.

- Sense pool, Model 2 (alternative): in the three codes with context-dependent stops (Tables 27, 28, 31), codons that function as amino acids in mid-sequence and as stops at gene termini are treated as sense codons. For all other tables, Model 2 coincides with Model 1.

The model conventions are documented in the datasets accompanying `deficit_models_analysis.csv` and `keto_amino_balance_models.csv`.

Global alignment of the sense pool with the lattice. The first test records two quantities of the sense pool: the neutron deficit relative to a reference total, $\Delta_N = T_{\text{ref}} - N(\text{sense pool})$, and the proton residue $P(\text{sense pool}) \bmod 37$. For the standard code these are $\Delta_N = T(\text{Octet I}) - N(\text{All sense}) = 3700 - 3589 = 111 = 3 \cdot 37$ and $P(\text{All sense}) \equiv -1 \pmod{37}$ (Section 3.1.1); under Key 1 the first deficit is closed by the three stop codons to align $N(\text{All})$ with the invariant $T(\text{Octet I})$, and the proton residue by the start-codon parameter $p_{\text{start}} = 1$. For the neutron deficit, three model choices of T_{ref} are considered: the constant $T(\text{Octet I}) = 3700$ of the standard code, the total nucleon content of the eight structural Octet I families under the amino-acid assignments of the specific table, and the total nucleon content of all 4-fold degenerate codon families in the specific table. For the proton residue the relevant quantity is intrinsic to the sense pool and requires no reference total. The full set of computed values across all tables and model combinations is tabulated in `deficit_models_analysis.csv`.

The two records give parallel outcomes: in each, the arithmetic property holds in exactly the same three tables. Table 1 (the standard code) and Table 11 (the Bacterial, Archaeal and Plant Plastid Code) have $\Delta_N = 111 = 3 \cdot 37$ under all three reference models and $P(\text{sense pool}) \equiv -1 \pmod{37}$ under both sense-pool models; their amino acid assignments are identical and they share the same set of stop codons, so arithmetically they are the same code applied in different domains. Table 28 (the Condylostoma Nuclear Code) gives the same values as the standard code under Model 1 and different values under Model 2: its three context-dependent stop codons, once excluded alongside ATG, leave a sense codon set identical to the one used for the standard code, whereas under Model 2 they are returned to the pool and neither the deficit nor the residue matches. In the remaining 24 tables neither property holds under any model combination. That the two records identify the same tables reflects a single underlying sense pool rather than independent agreement: Tables 1, 11, and (under Model 1) 28 all share the standard code's sense-codon assignments.

Across the 27 tables, six reasonable definitions of the neutron deficit and two of the proton residue, only the standard code and its exact codon-to-amino-acid equivalent (Table 11) satisfy both conditions under every model. Tables that reassign codons within the structural Octet I families, shift the set of 4-fold-degenerate families, or use a different number of stop codons destroy the match between the sense pool and the lattice of step 37. The global alignment is therefore not a generic feature of genetic codes.

Preservation of the Trp–Ile localization. The second test asks whether the localization of the Keto/Amino asymmetry in the pair Trp–Ile survives in alternative codes. The relevant signature is the pair of sense-pool differences $N(\text{Keto sense}) - N(\text{Amino sense}) = 37$ and $P(\text{Keto sense}) - P(\text{Amino sense}) = 36$, the $(37, 36)$ signature; the values across all tables are computed in `keto_amino_balance_models.csv`. The signature is reproduced exactly in seven tables. Five—Table 1 (Standard), Table 6 (Ciliate, Dasycladacean and Hexamita Nuclear), Table 11 (Bacterial, Archaeal and Plant Plastid), Table 29 (Mesodinium Nuclear), and Table 30 (Peritrich Nuclear)—reproduce it under both sense-pool models; the remaining two, Table 27 (Karyorelict Nuclear) and Table 28 (Condylostoma Nuclear), reproduce it only under the primary interpretation of context-dependent stop codons.

The mechanism is structural and operates differently in different tables. Table 11 has an amino acid assignment identical to the standard code and shares its set of stop codons, so the signature holds trivially. In Tables 6, 29, and 30 the codons TAG and TAA are reassigned to the same amino acid (Gln, Tyr, or Glu respectively); since TAG sits in the Keto branch of family TA and TAA in the Amino branch, this symmetric reassignment contributes identically to both branches of the sense pool and the Keto/Amino differences are preserved. No codons other than TAA and TAG are reassigned in these tables; in particular, neither TGG (Trp) nor any codon of Ile is changed, and the Trp/Ile localization of Section 3.1.5 continues to hold. Tables 27 and 28 involve context-dependent stop codons and behave differently under the two models: Table 28 declares all three of TAA, TAG, TGA as context-dependent stops, so under Model 1 its sense pool is identical to that of the standard code and the signature is reproduced, while under Model 2 the three codons re-enter the pool and it is lost; Table 27 permanently reassigns TAA and TAG to Gln (the same symmetric mechanism) and treats only TGA as a context-dependent stop, so Model 1 preserves the signature while Model 2, including TGA as Trp in the Keto branch, breaks it. Tables that reassign ATA (typically to Met in mitochondrial codes) or TGA (typically from stop to Trp) change the amino acid content of the carrier codons and no longer reproduce the signature.

The asymmetry between nuclear and mitochondrial codes is informative. All seven tables matching the $(37, 36)$ signature are nuclear or bacterial; none of the thirteen mitochondrial translation tables in the NCBI registry reproduces it, the failure tracing in each case to the reassignment of ATA or TGA, both altering codons on which the localization depends.

Relation between the two tests. A finer decomposition connects the seven matching tables to the first test. In the standard code the four sense-pool half-quantities have the residues $(N_{\text{Keto}}, N_{\text{Amino}}, P_{\text{Keto}}, P_{\text{Amino}}) \equiv (0, 0, -1, 0) \pmod{37}$, an arithmetic profile combining divisibility of each half-pool by 37 with a one-unit proton residue: parity bars the proton half-pools from the odd multiples of 37, and the specific nucleon counts place the residue at exactly one. Tables 1, 11, and 28 (under Model 1) reproduce all four residues exactly—for Table 11 by identity with the standard code, for Table 28 under Model 1 because its sense pool coincides with the standard once the context-dependent stops are excluded—and these are exactly the three tables that match in the global-alignment test above. The four additional tables matching the $(37, 36)$ signature (6, 27, 29, 30) show a different profile: N_{Keto} and N_{Amino} share the same non-zero residue mod 37, and P_{Keto} and P_{Amino} differ by one unit, but the absolute level has detached from the lattice. This is the signature of symmetric reassignment of TAA and TAG to a common amino acid: each half-pool acquires the same non-zero contribution, so the Keto/Amino differences are preserved while the divisibility of each half by 37 is lost. Among the 27 tables, 13 preserve the divisibility of at least one half-pool by 37, three of them mitochondrial (Tables 4, 16, and 23).

The global-alignment test measures the absolute alignment of the whole sense pool with the 37-lattice and matches only tables arithmetically equivalent to the standard code in their sense pool (Tables 1, 11, and 28 under Model 1). The localization test measures the preservation of a relational pattern between the Keto and Amino branches and additionally matches tables in which the only divergence from the standard code is a symmetric reassignment of TAA and TAG (Tables 6, 27, 29, 30). Together they show that the parametrization-independent sense-pool arithmetic of the standard code, in both its absolute and its relational form, is specific to the standard code among the known natural codes, holding elsewhere only in tables that are arithmetically equivalent to it or differ from it by a symmetric reassignment of TAA and TAG.

4.2 Specificity of the third codon position, the modulus 37, and the molecular representation

Two features are built into the framework of this work rather than derived from the data: the three chemical axes are read off the third codon position (Section 2.3), and divisibility by 37 is the primary criterion (Section 2.5). Two deterministic controls test whether the observed structure is specific to these two choices or would arise generically.

Codon position. Applying the same three chemical axes to the first and second codon position, instead of the third, removes the structure almost entirely. At the third position the Amino and Weak groups have all four nucleon quantities divisible by 37, the Keto and Strong groups have their neutron pool divisible, and the Pyrimidine group has Δ divisible. At the first position the only divisibility is the neutron pool of the Keto/Amino axis, present in both its complementary halves, and this is not an independent fact: Keto and Amino are complementary, so their sense-pool neutron counts sum to $N(\text{All sense}) = 3589 = 97 \cdot 37$ (Section 3.3.1), and divisibility of one half forces divisibility of the other. At the second position no nucleon quantity of any axis is divisible by 37. The control thus shows that divisibility by 37 along the chemical axes is confined to the third codon position: the same axes applied to the first and second positions do not meet the 37-lattice, the single first-position exception being the dependent one noted above. This excludes the alternative in which any equal-size split of the sense pool into chemical halves would align with the 37-lattice irrespective of position. The full values for all three positions are tabulated in `position_axis_analysis.csv`.

Choice of modulus. For every prime up to 97 we count how many of the four nucleon quantities of the codon groups it divides, over three value sets: the parametrization-independent groups (those with no service codon, on which 37 cannot be imposed), the groups under Key 0 (which does not impose 37), and the groups under Key 1 (whose derivation does). The scan ranges over primes rather than all integers because divisibility by a composite is the conjunction of divisibilities by its prime-power factors and introduces no new modulus, and because raising a prime to a higher power lowers the chance rate $1/m$ and thereby inflates the excess over the same quantities without adding information; the prime is the appropriate unit. The informative statistic is accordingly the excess over the chance rate $1/p$, computed over distinct values so that quantities forced to coincide by the structure of the code are not counted more than once.

Three primes divide their share at essentially the chance rate (3, 7 and 13). Two exceed it modestly, by less than a factor of three across the three value sets: 2, because every canonical amino acid has an even proton count (Section 3.1.5), and 5, reflecting the round-number regularity already reported in Section 3.5, which the scan thus corroborates. Against this background 37 divides five of the twenty-eight distinct parametrization-independent quantities, thirteen of the seventy distinct Key 0 quantities, and ten of the forty-seven under Key 1 — in each case several times the naive rate 1/37. This excess ratio is descriptive, not a p-value: the scan is deterministic and has no null model, and the nested group hierarchy makes the quantities non-independent, so the ratio alone cannot be read as a significance. An additional descriptive distinction not captured by that ratio is that the multiples of 37 contain several distinct odd cores, whereas the comparable factors 73 and 61 each reduce to a single core. Because the effect is already present in the value sets where 37 is not imposed, it is in any case not an artifact of the Key 1 derivation.

Three features set 37 apart from the two small primes, all independent of the number base. It is a larger prime, so each divisible quantity is rarer under the null. It has no structural counterpart to proton parity. And, unlike 5 — whose multiples in these value sets are almost all multiples of 25, so that the larger excess one would attribute to 25, 50 or 100 only re-scores the same quantities against a smaller chance rate — the modulus 37 occurs at a single power: no quantity is divisible by 37^2, so its excess does not grow when the modulus is enlarged. Reducing each quantity to its odd part — dividing out the powers of two that the nested group hierarchy (Octet I and its sixteen- and eight-codon subgroups) introduces mechanically, since a group of size $2k$ has twice the nucleon counts of its size-k refinement — sharpens this. Among the Key 0 quantities, 37 divides ten distinct odd cores; the multiples of $73 = 2 \cdot 37 - 1$ (the value of Δ for the Octet II eight-codon groups) collapse to that single core, and the multiples of 61 to the single core $549 = 9 \cdot 61$, to which the pyrimidine proton pool $P(\text{Pyrimidine}) = 2196 = 2^2 \cdot 3^2 \cdot 61 = 4 \cdot 549$ and the proton count of each pyrimidine letter $1098 = 2 \cdot 549$ both reduce. The factor 73 is thus an echo of the modulus itself; the factor 61 is an isolated factorization unrelated to 37 (its quantities deviate from the nearest multiples of 37 by $+13$ and -12, not by ± 1 or ± 2); and the factor 5 reduces entirely to 10 and 25, every quantity here divisible by 5 being divisible by 10 or by 25. The odd-core counts are tabulated in the corresponding column of `prime_divisibility_scan.csv`.

Representation dependence and the invariant relational core. The nucleon counts throughout are those of the free (neutral) amino acids — the canonical molecular identity of the species each codon encodes — under the most-abundant-isotope convention (^{1}H, ^{12}C, ^{14}N, ^{16}O, ^{32}S). Any alternative representation of the encoded amino acid differs from this one by a fixed nucleon offset per residue: the peptide residue removes one water (10 protons, 8 neutrons), the zwitterion removes nothing. Under such a uniform offset $(\delta P, \delta N)$, a linear combination of the counts is invariant if and only if its signed codon coefficients sum to zero; the control `representation_sensitivity.csv` records every anchor under all three representations. This splits the regularities in two. The relational regularities — the neutron and proton differences of the localizing pair Trp–Ile ($|\delta N| = 37$, $|\delta P| = 36$) and the Keto/Amino and Strong/Weak axis differences (37, 36) — are differences between codon sets of equal cardinality and are therefore invariant under every such re-representation. The absolute regularities — the sense-pool sums $N(\text{All sense}) = 97{\cdot}37$ and $P(\text{All sense}) \equiv -1 \pmod{37}$, the deficit $111 = 3{\cdot}37$, and the divisibility of the individual axis half-pools — are not: a uniform offset preserves the neutron alignment only when $60\,\delta N \equiv 0 \pmod{37}$, i.e., $\delta N \equiv 0$ (since $\gcd(60, 37) = 1$), which among chemically meaningful representations selects the free amino acid alone; under the peptide residue the sums become $N = 3109 \equiv 1$ and $P = 3580 \equiv 28 \pmod{37}$. The absolute alignment is therefore a property of the standard code's assignment in the canonical free-amino-acid representation, fixed a priori by molecular convention rather than by tuning, while the relational structure — the load-bearing part of the analysis — is representation-independent.

The functional-group asymmetry across positions. One observation on the Keto/Amino axis reaches beyond the third position. The asymmetry $N(\text{Keto}) - N(\text{Amino})$ that localizes to the Trp–Ile pair at the third position (Section 4.1) equals 37 there; the same Keto/Amino axis — the $\{G, T\}$ versus $\{A, C\}$ split — yields a quantity divisible by 37 at the first position as well (N-asymmetry $-333 = -9 \cdot 37$) and at the second (T-asymmetry $185 = 5 \cdot 37$). The divisible quantity is N, T, N by position, fixed after inspection rather than in advance, so this is a single observation along one axis rather than a control of the kind above. The three are arithmetically independent: each is a sum over a different partition of the same sixty codons, and the divisibility at one position neither follows from nor implies it at another; what they share is only the choice of axis. Across the 27 NCBI translation tables this three-position coincidence is specific to the standard code and its exact codon-to-amino-acid equivalent (Table 11): no other table reaches more than one position, and the tables that preserve the Trp–Ile localization (Tables 6, 29 and 30) reproduce only the third-position value. Three caveats further qualify it. Unlike the third-position case, the first- and second-position asymmetries do not localize to a single amino-acid pair — each receives contributions from nineteen of the twenty amino acids, because the cancellation that isolates Trp and Ile relies on third-position synonymy, which the other positions lack. Their non-divisible components carry no analogue of the parity-induced residue -1 that accompanies the third-position value. And, as in Section 4.1, the 27 tables are not statistically independent, so this is evidence of specificity among natural codes rather than of improbability under a null model. The per-table values are tabulated in `position_asymmetry_ncbi.csv`.

All three controls are deterministic and exact-integer, and are regenerated by `controls.py` (Section 2.6).

4.3 Significance under a random-code null

The controls of Section 4.2 show that divisibilities concentrate in third-position groupings and that modulus 37 stands out among the tested prime moduli; the comparison of Section 4.1 shows that, among the 27 NCBI translation tables, the tested combination occurs only in the standard code and its exact codon-to-amino-acid equivalent. Specificity among realized codes, however, is not improbability under an explicit null. We therefore test the standard code against a random-code null of the kind used in studies of genetic-code optimality [2]: the architecture of the code is held fixed (the 64 codons, the synonymous blocks with their sizes, the three stop codons, the Rumer partition, and the third-position axes), and the only thing randomized is which amino acid, carrying its fixed proton and neutron counts, occupies each of the twenty blocks. A trial is a uniformly random permutation of the twenty real paired proton and neutron profiles over the twenty synonymous blocks. The modulus 37 is taken a priori from the prior literature; the statistical summaries were formally defined before the final Monte Carlo run. Because the analysis was not preregistered before inspection of the observed structure, the results are interpreted as exploratory.

S1–S3 and S4sense are evaluated over the sixty-codon sense pool without assigning any values to service codons. S4sense averages the distances to the nearest multiple of 37 for the four quantities T, P, N, and Δ over all 33 groups, for 132 distances in total. Under S4a the stop codons contribute zero, while the contribution at ATG in each random code is the profile of the amino acid assigned by the permutation to the ATG block. Under S4b each random code receives its own minimal nonnegative Key 1, re-derived by the same Keto/Amino balance and global divisibility conditions used for the standard code.

For every statistic the Monte Carlo tail probability is calculated as $p_{\mathrm{MC}} = (b + 1)/(M + 1)$, where b is the number of random codes at least as extreme as the standard code and $M = 10^6$. A 95% Monte Carlo uncertainty interval is reported with each estimate. The computation is reported in `mc_significance.py` and summarized in `mc_significance.csv`. An independent implementation, using a different construction of the codon groups, a direct search for Key 1, and a different random seed, reproduces the observed statistics and tail probabilities within the expected Monte Carlo error (`verify_mc.py`).

Six statistical summaries are considered. S1 tests the joint divisibility of two anchors: $N(\text{All sense}) = 3589 = 97 \cdot 37$ and $T(\text{Octet I}) = 3700 = 100 \cdot 37$. Both events occur in a fraction 0.000800 of random codes (Monte Carlo 95% interval 0.000746–0.000856). This is the directly estimated joint probability; no assumption that the two events are independent is used in its calculation.

S2 counts divisibility by 37 among the four quantities T, P, N, and Δ for nine sense-pool groups: the full pool, the six chemical-axis half-pools, and the two octets. The standard code has 13 divisible quantities among 36 nominal entries, against a null mean of 1.0299. A value of 13 or greater occurred in 39 of 10^6 trials, giving $p_{\mathrm{MC}} = 4.0 \times 10^{-5}$ with a 95% interval of 2.9×10^{-5}–5.3×10^{-5}. The components of the count are dependent: $T = P + N$, $\Delta = P - N$, and the Keto/Strong and Amino/Weak groups form exact twin pairs. These dependencies are reproduced in every permutation and are therefore already included in the null distribution of the composite statistic.

S3 applies the same count to the 17 groups that contain no service codon at all, giving 68 nominal entries. The standard code has 14 divisible quantities against a null mean of 2.1311, $p_{\mathrm{MC}} = 0.025135$, and a 95% interval of 0.024829–0.025443. Eleven of the fourteen divisibilities are structural copies of $T(\text{Octet I}) = 100 \cdot 37$, while the remaining three form the pyrimidine Δ-cascade ($\Delta(\text{Pyrimidine}) = 8 \cdot 37$ and $\Delta(\{C\}) = \Delta(\{T\}) = 4 \cdot 37$). S3 therefore records the propagation of two structural cascades and is treated as a secondary summary.

S4sense measures not only exact divisibility but the overall proximity of the sense pool to the lattice of step 37. For each of the 33 groups, the distances of T, P, N, and Δ to the nearest multiple of 37 are averaged, with zero included as a distance of zero; 132 quantities are included in total. The standard code has mean distance 4.8182 against a null mean of 9.1495. An equal or smaller distance occurs in a fraction 0.000274 of random codes, with a 95% interval of 0.000242–0.000307. This statistic assigns no values to service codons. Because the summary was defined after inspection of the observed structure, it is treated as an exploratory continuous sensitivity analysis.

S4a applies the same distance under Key 0. In the standard code ATG encodes methionine; in each random code its contribution is the amino acid actually assigned by the permutation to the ATG block, while the stop codons contribute zero. The standard code has mean distance 5.5606 against a null mean of 9.0157, $p_{\mathrm{MC}} = 0.002807$, and a 95% interval of 0.002704–0.002912. The concentration is therefore already present under the naive Key 0, although Key 0 remains a formal assignment of service-codon values.

S4b tests the Key 1 derivation symmetrically: the standard code and every random code are evaluated under their own minimal nonnegative key, derived from the Keto/Amino balance and global divisibility by 37. The fixation of this axis and the compatibility of the conditions are established in Section 3.3.3. The standard code has mean distance 4.6364 against a null mean of 7.7397, $p_{\mathrm{MC}} = 0.021106$, and a 95% interval of 0.020825–0.021389. Thus the standard code's Key 1 is moderately atypical among keys re-derived for random codes.

The two most sensitive sense-pool summaries — the discrete S2 and the continuous S4sense — describe the same overall concentration from different directions. Their p-values are dependent and must not be multiplied or interpreted as two independent lines of evidence. S1 records two individual anchors of the structure, whereas the weaker S3 shows that restriction to groups with no service positions leaves mainly the Octet I and pyrimidine-Δ cascades.

S4a and S4b address separate questions about the two formal assignments to service codons. S4a shows that the concentration is retained under the naive Key 0. S4b evaluates the entire Key 1 derivation rather than the fixed values of the standard key and gives a more moderate tail probability. The assignment-free sense-pool results therefore form the central statistical part of the analysis, while S4a and S4b are secondary tests of its extension to the complete table.

All reported p_{MC} values are nominal tail probabilities for the individual statistics; no multiplicity correction across the six statistics was applied.

The null model fixes the standard code's degeneracy architecture and the real paired proton and neutron counts of the amino acids and randomizes only their assignment to blocks. A different generative model would give different probabilities. The modulus 37, the Rumer partition, and the chemical axes have substantive a priori origins, but the choice of global statistical summaries was not preregistered before inspection of the structure. The results therefore characterize the exploratory atypicality of the standard code under one explicit combinatorial model rather than confirming a preregistered hypothesis. Improbability under this null concerns combinatorial codes, a different question from specificity among the 27 natural codes in Section 4.1; it speaks to no biological mechanism.

4.4 Relation to prior work

Three prior contributions are particularly relevant to the arithmetic structure analyzed in this work. Yu. B. Rumer [6, 7] introduced the partition of the 64 codons of the standard code into two equal classes: Octet I consisting of the eight fully degenerate families, and Octet II consisting of the eight heterogeneous families. The two classes are related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$, under which each family of one class is mapped to a family of the other. Shcherbak and Makukov [8] observed that the number 37 plays a recurrent role in the genetic code, appearing as a divisor of amino acid nucleon sums across various groupings of codons. Panov and Filatov [5] revisited and extended this analysis. They introduced the per-codon counting rule, in which each of the 64 codons contributes independently the nucleon count of the amino acid it encodes. They also proposed that stop codons be assigned a total nucleon count of 74 each rather than zero.

The present work differs from these in several methodological choices. Shcherbak and Makukov, and Panov and Filatov following them, decompose each amino acid into a constant part with nucleon count 74 and a variable side chain, and many of their nucleon sums are formed over side chains alone. This decomposition forces a special treatment of proline, whose cyclic side chain reduces the nucleon count of its constant part to 73 rather than 74: to bring it in line with the other nineteen amino acids, Shcherbak and Makukov introduce a virtual transfer of one nucleon from its side chain to its constant part, an operation they call the *activation key*. The present work uses full amino acid nucleon counts without separation into constant and side-chain components, and the question of how to treat proline does not arise. Shcherbak and Makukov also adopt a counting rule in which each amino acid contributes once per codon family rather than once per codon; Panov and Filatov replace this with the per-codon rule, which the present work also uses. Finally, all nucleon sums in the prior works are formed over the total nucleon count T (or over its restriction to side chains); the separation of T into protons P and neutrons N, and the analysis of the difference $\Delta = P - N$, are introduced in the present work and underlie a class of arguments unavailable when only T is considered.

The two prior works also differ from ours in the systematics by which codons are grouped. They apply several different grouping schemes in parallel (Gamow's sorting and others), each tailored to a particular set of patterns. The present work formulates a single grouping system based on the Rumer octets and the third-position chemical axes, and analyzes the arithmetic structure within it consistently. The parametrization of service codons also differs: Shcherbak and Makukov treat stop codons as contributing zero and identify ATG with Met throughout, Panov and Filatov propose in addition the assignment of total nucleon count 74 to stop codons as an alternative, while the present work considers the parametrization of service codons as part of the derivation of Key 1 (Section 3.3). Finally, Shcherbak and Makukov work extensively with the euplotid code variant (in which TGA is reassigned to Cys), which they take as a symmetrized counterpart of the standard code; Panov and Filatov treat this as a switchable alternative. The present work focuses on the standard code, and compares its arithmetic structure with the 26 alternative NCBI translation tables on equal footing (Sections 4.1, 4.1).

We now describe how the divisibility findings of these prior works project onto the groups defined in the present analysis. Shcherbak and Makukov, counting one molecule per distinct coding role within a family, observed that the sum of amino acid nucleon counts over Octet I is divisible by 37, with $T(\text{Octet I}) = 925 = 25 \cdot 37$, and that the same holds for Octet II under the assignment of zero to stop codons, with $T(\text{Octet II}) = 2220 = 60 \cdot 37$. Panov and Filatov, having introduced the per-codon counting rule, obtained the corresponding totals under codon-level counting: $T(\text{Octet I}) = 3700 = 100 \cdot 37$ and $T(\text{Octet II}) = 4218 = 114 \cdot 37$ (or $4440 = 120 \cdot 37$ under their alternative assignment of 74 to stop codons).

The relation between the per-coding-role total and the per-codon total is direct for Octet I, which is fully four-fold degenerate: $3700 = 4 \cdot 925$. For Octet II, whose families contain different numbers of distinct coding roles, the per-codon total 4218 is not a multiple of the per-coding-role total 2220 and is an independent fact. The per-coding-role total 2220 itself coincides with $T(\text{Keto} \cap \text{Octet II}) = T(\text{Strong} \cap \text{Octet II})$ in our analysis: the universal pyrimidine synonymy within Octet II (Section 3.1.3) implies that the sixteen codons of Keto (or of Strong) intersected with Octet II contain each coding role exactly once, so their per-codon total equals the per-coding-role total.

From these primary facts, the divisibility by 37 of several further groups in our systematics follows trivially. The total over the entire code is the sum of the two octet totals; the sixteen-codon and eight-codon subgroups of Octet I are mutually identical by four-fold degeneracy; and the four chemical-axis groups Keto, Strong, Amino, Weak each decompose into an Octet I sixteen-codon subgroup and an Octet II sixteen-codon subgroup, of which the first is automatically divisible by 37 and the second reduces to a separate question addressed in the present work.

4.5 Limitations and outlook

The statistical significance of the observed arithmetic regularities is assessed in Section 4.3 under one explicit null — the random-code permutation, which fixes the code's degeneracy architecture and the amino-acid nucleon counts and randomizes only the assignment of amino acids to codon families. Under that null the headline divisibility count and the continuous concentration over the full sense pool are highly atypical. The narrower S3 test, restricted to nested groups that contain no service codon at all, is only weakly atypical ($p_{\mathrm{MC}} \approx 0.025$) and mainly reflects copies of the Octet I total. The choice of null is not unique: alternative generative models, for example randomizing the block sizes themselves or the nucleon counts, would probe different aspects of the regularity and are left for future work, as is any account of a mechanism that could produce it.

The parametrization of service codons in the present work is restricted to Key 0 and Key 1; additional parametric assignments are tabulated in `key_parameters.csv` and in the accompanying `visualization.html` for illustration of other choices.

A further direction concerns the group-theoretic structure of the codon table. Following Panov and Filatov [5], to each of the 24 orderings of the four bases on the axes of the codon matrix we associate its *calligram*: the geometric pattern formed by the eight homogeneous codon families of Octet I on the 4×4 matrix. Connectedness of an octet refers to the connectedness of this pattern under 4-adjacency of matrix cells. Among the 24 calligrams, Panov and Filatov [5] identified four — CTGA, CGTA, ATGC, and AGTC — in which each of the Rumer octets forms a connected region, and showed that these four form a closed set under the transformations $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$, $R_1 = (\mathrm{T} \leftrightarrow \mathrm{G})$, and $R_2 = (\mathrm{C} \leftrightarrow \mathrm{A})$. These three transformations together with the identity form the Klein four-group V_4, and the action of V_4 extends naturally to the full set of 24 calligrams. This action of V_4 on the full set of 24 calligrams is free, partitioning it into six orbits of four calligrams each. Exactly one of these six orbits consists of connected calligrams, accounting for the four calligrams above; the remaining five orbits contain no connected calligram. This connected orbit furthermore contains the canonical chemical ordering C, G, T, A of Rumer [7] (Section 2.2); the correspondence between this chemical ordering and the geometric property of connectedness is non-automatic, since only two of the eight chemically alternating orderings of the four bases yield connected octets. The search for further structural invariants distinguishing the orbits — and for connections between the group-theoretic orbits and the arithmetic systematics of the present work — is left to future investigation.

5 Conclusion

The standard genetic code exhibits arithmetic regularities visible at several levels of parametric specificity. A first layer is independent of any choice of nucleon contributions for the service codons, and it carries the central finding of this work: summed over the sixty sense codons, the neutron pool is an exact multiple of the modulus, $N(\text{All sense}) = 3589 = 97 \cdot 37$, while the proton pool falls one unit short of an odd multiple, $P(\text{All sense}) \equiv -1 \pmod{37}$—a property of the proton/neutron split that is invisible at the level of the total count T. Two further parametrization-independent facts sit alongside it. The eight families of Octet I carry $T(\text{Octet I}) = 3700 = 100 \cdot 37$, with $P(\text{Octet I}) = 2000$ and $N(\text{Octet I}) = 1700$, the latter combining with the Octet II sense pool as $1700 + 1889 = 3589$; and the entire Keto/Amino asymmetry of the code localizes, after exclusion of the service codons, to the single amino-acid pair Trp–Ile, with $|\delta N| = 37$ and $|\delta P| = 36$. The universal pyrimidine synonymy within Octet II produces identities between the twin sixteen-codon groups Keto/Strong and Amino/Weak within Octet II.

A second layer arises under the naive reading, where the proton and neutron counts P and N are analyzed separately. The total nucleon count T is divisible by 37 for the four chemical-axis groups and for Octet II. For the Amino and Weak groups, P and N are individually divisible by 37, aligning these groups with the lattice of step 37 along both axes.

A third layer introduces the service-codon assignment. There is a unique minimal non-negative integer assignment, Key 1 (Section 3.3), under which the parametrization-independent structure extends to the full 64-codon table. Key 1 is not a foundation for the findings above but a device that renders them on the complete table: the same sense-pool divisibility, the same localized pair, and the same proton parity that hold with the service codons unassigned are exactly what make the assignment solvable in small integers. The neutron balance across the Keto/Amino axis reproduces the Trp–Ile difference $|\delta N| = 37$; the global neutron deficit $T(\text{Octet I}) - N(\text{All sense}) = 111 = 3 \cdot 37$ closes because it is a common multiple of the number of stops and the modulus; the even-proton parity theorem forbids $|\delta P| = 37$ and bars the sense-pool proton pool from the odd multiples of the modulus, so that it sits one unit below—by a margin that the specific nucleon counts make exactly one—which the start codon's minimal contribution $p_{\text{start}} = 1$ restores; and $\gcd(4, 37) = 1$ leaves a unique minimal solution. None of these is a parametric assumption; each is an empirical property of the standard code established at the parametrization-independent level, and Key 1 exists as the minimal extension precisely because they hold simultaneously.

Under the derived parametrization, the arithmetic regularities organize into a connected family: alignment of the four chemical-axis groups along the axes T, P, N, and Δ (Section 3.4.1), macroscopic identities including $N(\text{All}) = T(\text{Octet I}) = 3700$ and $P(\text{All}) = T(\text{Octet II}) = 4292$ (Section 3.4.2), the uniform invariant $\Delta = 296$ across all six 32-codon groups (Section 3.4.4), and divisibility patterns by 37 and 999 at multiple levels of partitioning (Section 3.4.3). The comparative analysis across the 27 NCBI translation tables (Section 4.1) shows that the simultaneous holding of all structural preconditions enumerated above is a property of the standard code alone.

Two features of this structure constrain how it may be read. First, the regularities reside in group sums rather than in individual molecules: no single canonical amino acid has a nucleon count divisible by 37, and in groups of variable coding degeneracy, such as Amino, the divisibility holds only when codons are counted with their multiplicity and fails when each amino acid is counted once. The nucleon composition and the degeneracy of coding are therefore not independent quantities within this structure: the regularity is a joint property of how nucleon mass is distributed and how many codons each amino acid receives. Second, the entire first layer above, including the values over Octet I and the twin-group identities, is established with the service codons carrying no nucleon values at all; the derived parametrization of the service codons enters only at the later layers and does not affect the parameter-independent results. The assigned values are thus a device for exhibiting the arithmetic of the complete table, not a foundation on which the principal findings rest.

The specificity established by the comparison across natural codes is complemented by deterministic controls and by a Monte Carlo analysis. The controls show that divisibilities concentrate in third-position groupings and that 37 stands out among the tested prime moduli. Under an explicit random-code null that fixes the degeneracy architecture and permutes the twenty real paired proton and neutron profiles over the twenty blocks, two dependent sense-pool summaries place the standard code in the far tail: the discrete S2 count gives $p_{\text{MC}} = 4.0 \times 10^{-5}$, and the S4sense mean distance gives $p_{\text{MC}} = 2.74 \times 10^{-4}$. The narrower S3 test, restricted to groups that contain no service codon at all, is only weakly atypical ($p_{\text{MC}} \approx 0.025$) and mainly reflects dependent copies of the divisibility of $T(\text{Octet I})$. The concentration persists under Key 0 ($p_{\text{MC}} \approx 0.0028$), whereas the symmetric test of the Key 1 derivation gives moderate atypicality ($p_{\text{MC}} \approx 0.021$). These probabilities refer to different, dependent summaries and are not combined by multiplication. This is improbability under one combinatorial null, distinct from the specificity among the 27 natural codes, and it speaks to no biological mechanism; alternative null models are left for future work.

The methodological choice underlying the present work is to restrict attention to parameters whose definition is unambiguous — the total nucleon count T, its decomposition into protons P and neutrons N, and the difference $\Delta = P - N$ — and to a single grouping system based on the Rumer octets and the third-position chemical axes. The interpretation of the arithmetic facts collected here is left open.

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